# Gerontology / Medical Surgical Nursing 3

Dr Mazen AbuQamar



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  - The first true geriatric nurse
  - Nurse superintendent comparable to our current nursing homes

- Doreen Norton
  - 1956 speech at the annual conference of the SNA in London

# Landmarks

- 1950 – First geriatric textbook published
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# Roles of the Gero Nurse

- Provider of care
- Teacher
- Manager
- Advocate
- Research Consumer

# Gerontological Nursing Defined

- Until the middle of the 19 century, only two stages of human growth and development were identified: childhood and adulthood.

- The term geriatric comes from the Greek words “geras,” meaning old age, and “iatro,” meaning relating to medical treatment.
- Geriatrics is the medical specialty that deals with the physiology of aging and with the diagnosis and treatment of diseases affecting them. It focuses on abnormal conditions and the medical treatment of these conditions.

# Gerontological Nursing Defined

- Gerontology comes from the Greek words “gero,” meaning related to old age, and “ology,” meaning the study of. Thus, gerontology is the study of all aspects of the aging process, including the clinical, psychologic, economic, and sociologic problems of older adults and the consequences of these problems for older adults and society.

- The term gerontics “gerontic nursing”, was coined by Gunter and Estes in 1979 to define the nursing care and the service provided to older adults.
- Gerontic nursing encompasses a holistic view of aging with the goal of increasing health, providing comfort, and caring for older adult needs.

# What is Aging ?

**Table 1-1** Categorizing the Aging Population

| AGE (YEARS) | CATEGORY |
|---|---|
| 55 to 64 | Older |
| 65 to 74 | Elderly |
| 75 to 84 | Aged |
| 85 and older | Extremely aged |
| Or | |
| 60 to 74 | Young-old |
| 75 to 84 | Middle-old |
| 85 and older | Old-old |

**. Chronological age**
How long have you lived?"

**.Biological age:** "How old is your body?“
a complex biological processes in which changes at molecular, cellular, organ, and systemic levels result in a progressive, inevitable decrease in the body's ability to respond appropriately to internal and/or external stressors.

**Functional age:**
Age based on what physical activities an older adult can do on a day-to-day basis. E.g- bath, dress, prepare meals, use the phone, etc.

**. Psychological age:**
The behavioral capacities people use to adapt to changing environmental demands such as memory, motivation, learning ability, adaptation and coping with daily stressors, etc, maturity.

## GERONTOPHOBIA

- The fear of aging and the refusal to accept older adults into the mainstream of society.
- Gerontophobia sometimes results in very odd behavior. Teenagers buy antiwrinkle creams. Thirty-year-old women consider facelifts. Forty-year-old women have hair transplants.

## Ageism

- The extreme forms of gerontophobia are ageism and age discrimination. Ageism is the disliking of aging and older adults based on the belief that aging makes people unattractive, unintelligent, and unproductive.
- It is an emotional prejudice or discrimination against people based solely on age.
- Ageism allows the young to separate themselves physically and emotionally from the old and to view older adults as somehow having less human value.

# Chronic Disease

- Increases with age
- 4 leading causes of death
  - Heart disease
  - Stroke
  - Cancer
  - DM
- Sensory impairments and oral health problems more frequent

# When Does Aging Begin?

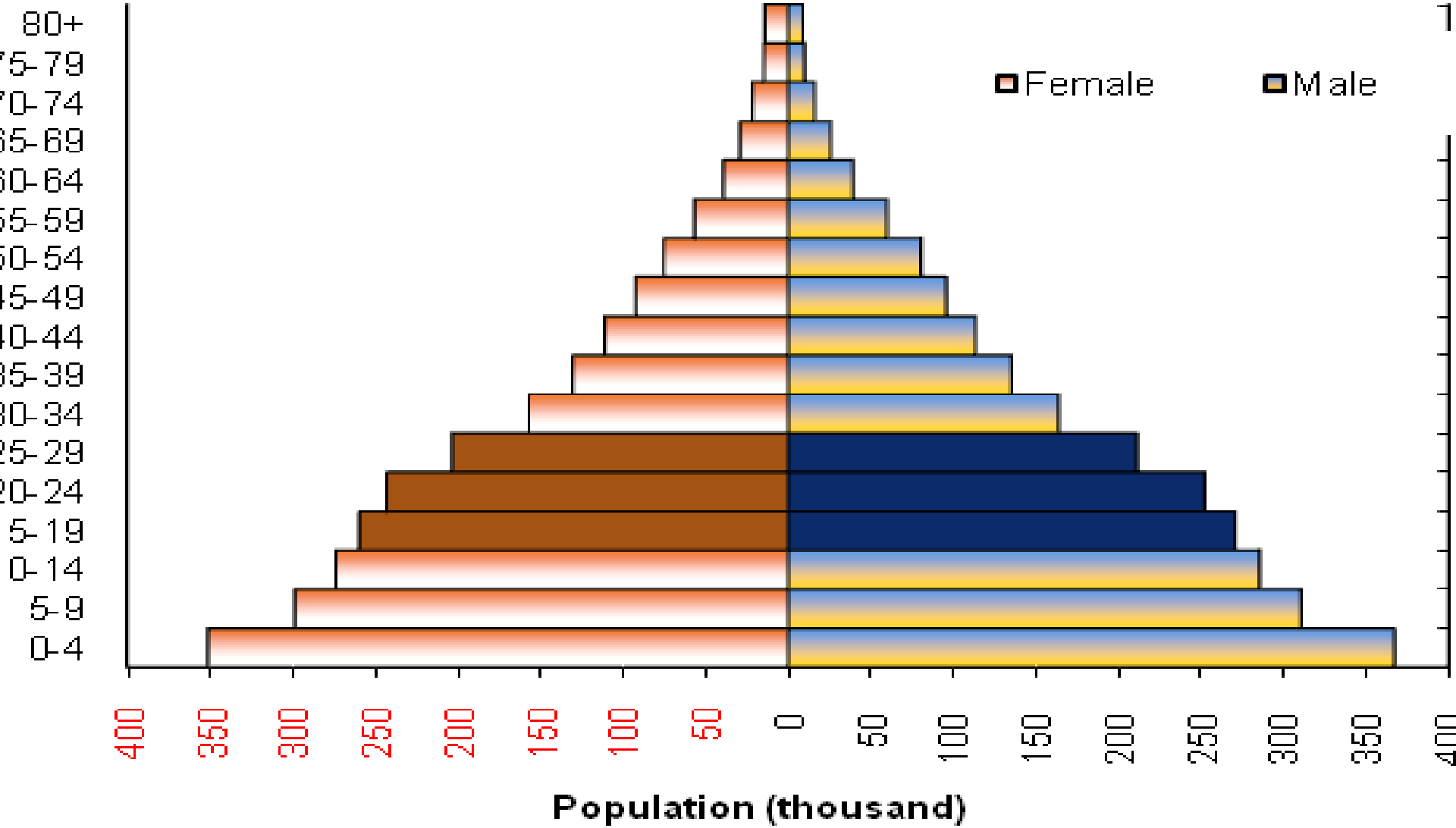
- Aging begins the day we are born
- No single measure of how “old” a person is
- Aging is highly individualized
- Aging proceeds at different rates in different people, and within different systems of the body

# Why Do People Age?

Many theories to include:

- Hereditary Factors
- Loss of cellular mass and ability of cells to divide and replicate
- Accumulation of waste materials that clog cells and cause them to die
- Changes in structure of connective tissue

No single theory adequately describes the aging process




Age distribution in Palestine 2016

# OLD AGE AND THE ELDERLY HAVE A VARIETY OF EXPECTATIONS AND NAMES. . . .FEW ARE POSITIVE

- **GEEZER**
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# Facts OF AGING

# FIVE FCTORS OF SUCCESSFUL AGING

- **LIFE SATISFACTION:** rewarding, few regrets, positive attitude about past and future
- **SOCIAL SUPPORT SYSTEM:** network of family and friends
- **GOOD PHYSICAL AND MENTAL HEALTH**
- **FINANCIAL SECURITY**
- **PERSONAL CONTROL OVER ONE'S LIFE:** independence, dignity, and self-worth

# A period of losses

- Loss from death (spouse, children, friends)
- Loss of work structure and income
- Loss of home and neighborhood
- Loss of social network
- Loss of primary control over decisions regarding health, mobility, living situation, finances

# Normal Changes of Aging

- Physical changes related to “Normal” aging ARE NOT disease
- Changes occur in most body systems to include:
  - Sensory System
  - Brain and Central Nervous System
  - Muscles and Bones
  - Digestion
  - Heart/Circulatory System
  - Respiratory System

# Physiological change

- Changes in body function with age are part of a continuum that starts the moment life begins.
- By approximately age 18, the human body reaches full anatomic and physiologic maturity.
- The peak years of physiologic function last from the late teens through the thirties—the so-called prime of life.
- Just as no two individuals grow and develop at exactly the same rate, no two individuals show the signs of aging at the same time.
- Heredity, environment, and health maintenance significantly affect the timing and magnitude of age-related changes.
- Some people are chronologically quite young but appear old.
- The most severe cases of this occur in a rare condition called progeria.



# five leading causes of death among older adults

- (1) heart disease,
- (2) cancer,
- (3) chronic lower respiratory disease,
- (4) cerebrovascular disease, and
- (5) Alzheimer disease.

# Theories of Aging

- There is no single universally accepted definition of *aging*.
- Aging is best looked at as a series of changes that occur over time, contribute to loss of function, and ultimately result in the death of a living organism.
- No one has identified a single unified rationale for why we age and why different people live lives of different lengths.
- Studies of families and identical twins show that there is a strong correlation in the life expectancies of genetically related people. If your grandparents and parents live to be 60, 70, 80, or 90 years of age, you are likely to have a similar life span.

# BIOLOGIC THEORIES

- Biologic theories of aging attempt to explain the physical changes of aging.
- Many biologic theories of aging overlap because most assume that the changes that cause aging occur at a cellular level (cell structures or function).
- **The programmed theory** proposes that everyone has a "biologic clock" that starts ticking at conception.
- In this theory, each individual has a genetic "program" that predetermined number of cell divisions (atrophy of the thymus, menopause, skin changes, and graying of the hair)
- **Run-out-of-program theory**, which proposes that every person has a limited amount of genetic material that will run out eventually
- **The gene theory** proposes the existence of one or more harmful genes that activate over time, resulting in the typical changes seen with aging and limiting the life span of the individual.

# BIOLOGIC THEORIES

- **The error theory proposes** that errors in ribonucleic acid protein synthesis cause errors to occur in cells in the body, resulting in a progressive decline in biologic function.
- **The somatic mutation theory** is similar but proposes that aging results from deoxyribonucleic acid (DNA) damage caused by exposure to chemicals or radiation
- **Cellular theories propose** that aging is a process that occurs because of cell damage. When enough cells are damaged, overall functioning of the body is decreased.
- **The free radical theory** provides one explanation for cell damage.
- Free radicals are unstable molecules produced by the body during the normal processes of respiration and metabolism or following exposure to radiation and pollution. These free radicals are suspected to cause damage to the cells, DNA, and the immune system.

# BIOLOGIC THEORIES

- **The wear-and-tear** theory presumes that the body is similar to a machine, which loses function when its parts wear out. As people age, their cells, tissues, and organs are damaged by internal or external stressors.

- **The neuroendocrine theory** focuses on the complicated chemical interactions set off by the hypothalamus of the brain.
- With age, the hypothalamus appears to be less precise in regulating endocrine function, leading to age-related changes such as decreased muscle mass, increased body fat, and changes in reproductive function. It is proposed that hormone supplements may be designed to delay or control age-related changes.

## Alternative and Complementary Therapies to Slow or Reverse Aging

- **Antioxidant Therapy** Proposed as a method of neutralizing free radicals, which may contribute to aging and disease processes
- Includes a number of vitamins and minerals, such as vitamins A, B6, B12, C, and E; beta carotene; folic acid; and selenium
- **HORMONE THERAPY** Proposed to replace a reduction in hormones, which naturally decrease with aging includes hormones, such as dehydroepiandrosterone (DHEA), estrogen, testosterone, melatonin, and human growth hormone (HGH)
- **SUPPLEMENTS** Proposed to replace or enhance nutritional status; often marketed as "natural" remedies include substances such as ginseng, coral calcium, Echinacea, and other herbal preparations
- **CALORIE-RESTRICTED DIET** Proposes that significant calorie reduction can extend life; based on studies in rats, mice, fish, and worms

## The immunologic theory

- **The immunologic theory** proposes that aging is a function of changes in the immune system.
- According to this theory, the immune system an important defense mechanism of the body weakens over time, making an aging person more susceptible to disease.
- The immunologic theory also proposes that the increase in autoimmune diseases and allergies seen with aging is caused by changes in the immune system.

# Types of Theories

- Psychosocial: Attempt to explain aging in terms of behavior, personality, and attitude change
- Encompass psychological and sociological theories
  - Psychological: How mental processes, emotions, attitudes, motivation, and personality influence adaptation to physical and social demands

# Gerontological Nursing Defined

- An encompassing view of the care of older adults (Mauk, pg. 5)
- Providing both physical and emotional support in addition to respectful care

- Florence Nightingale
  - The first true geriatric nurse
  - Nurse superintendent comparable to our current nursing homes

- Doreen Norton
  - 1956 speech at the annual conference of the SNA in London

# Advantages of Learning Geriatric Care

- Learning patience, tolerance, understanding, and basic nursing skills
- Witnessing the terminal stages of disease and the need for skilled nursing care
- Preparing for the future because the aged will always be a part of the care you provide
- Recognizing the importance of rehab
- A need for research

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# Old Defined

- Young old: 65 – 74
- Middle old: 75 – 84
- Old old: 85 and up
- Only a guide as there is a vast difference in biological and chronological aging

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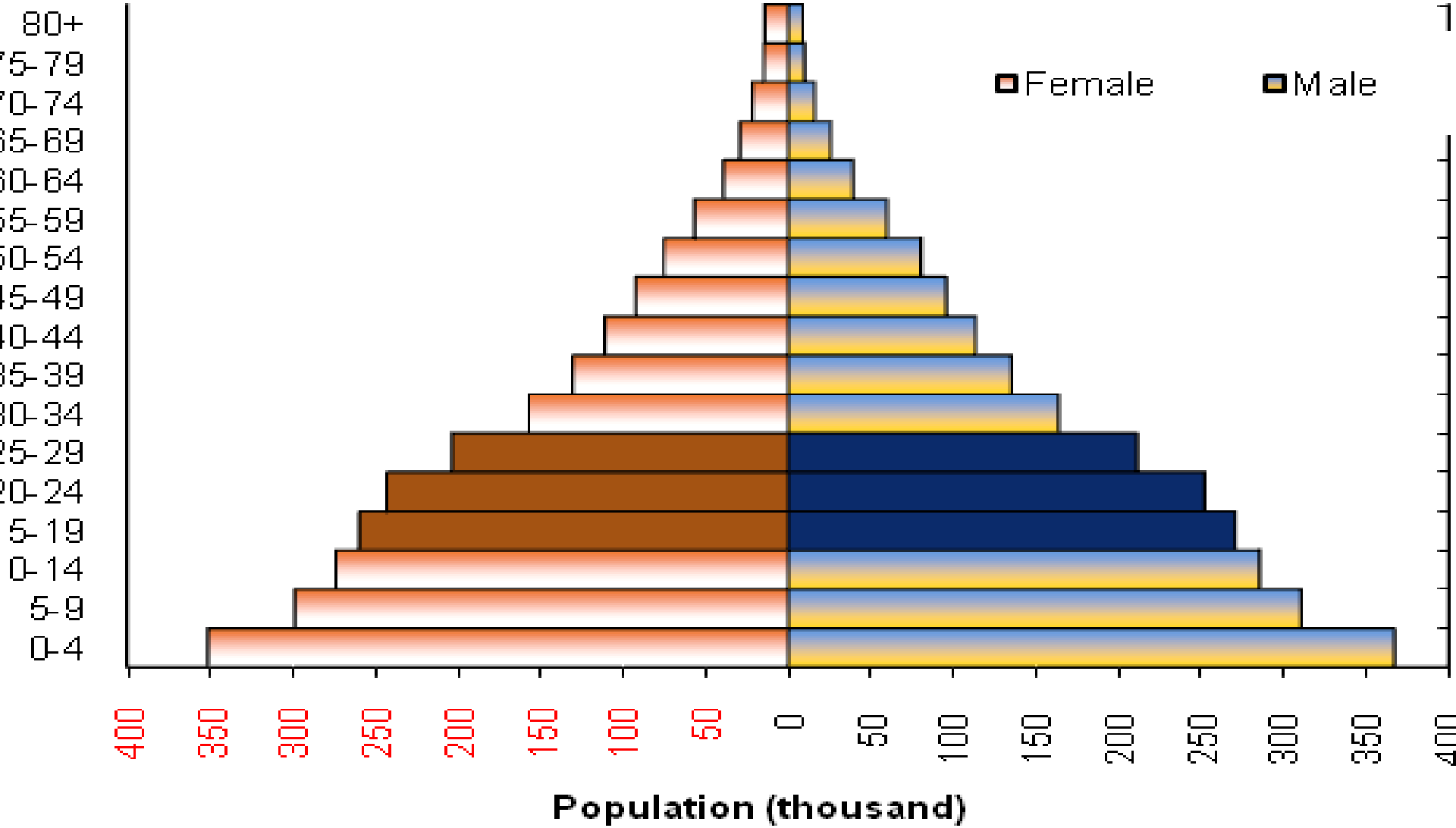
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Health Changes and Challenges



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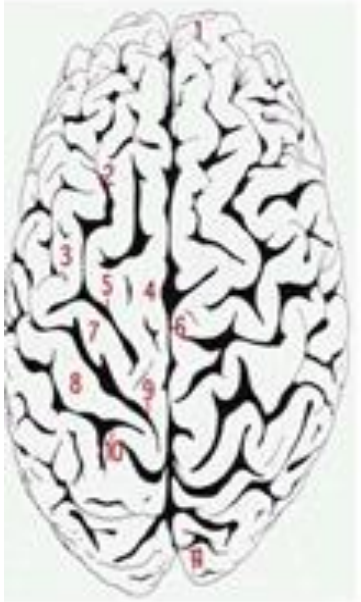
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# Facts OF AGING

- Age brings greater risk of disease
- Exercise cuts death rate 25 - 50% and osteoporosis less with strengthening exercies
- Reduction in mental sharpness
  - 20% of elderly suffer from Alzheimers
  - 50% have some cognitive decline
  - Decline often limited to storing new information
  - Performance, recognition, recall shows little decline
- Depression present, increase suicide rate in elderly
- Increase the concern in religious belief
- Old people are victims of crime
- Elderly most like to talk about the past



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# Alzheimer's Disease

- Difficulty with problem solving, complex tasks and sound judgments
- poor judgment and deepening confusion
- memory loss
- Undergo significant changes in personality and behavior
- Lose the ability to communicate coherently
- Require daily assistance with personal care

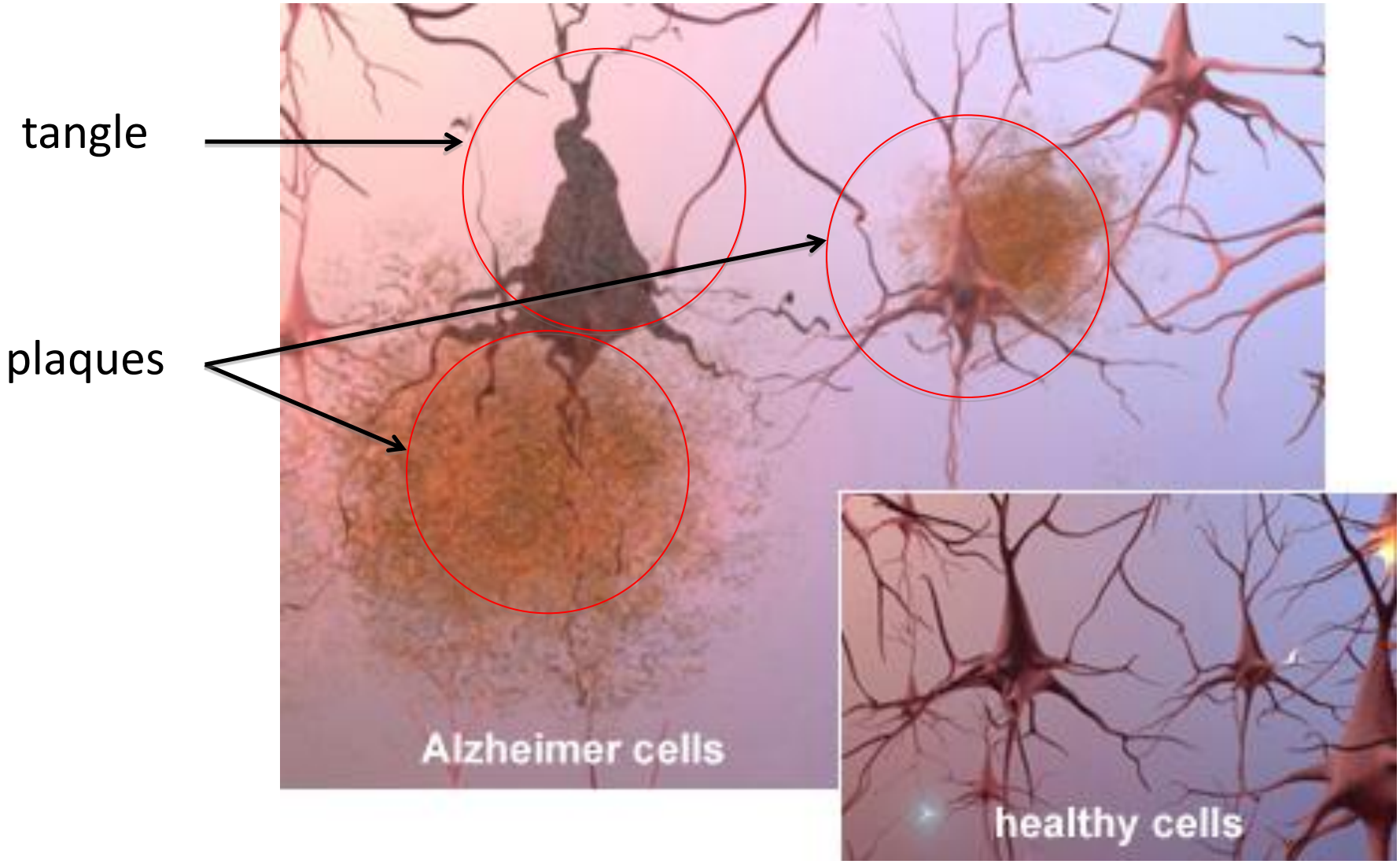
  total assistance with dressing, eating, etc.
- Experience a decline in physical abilities

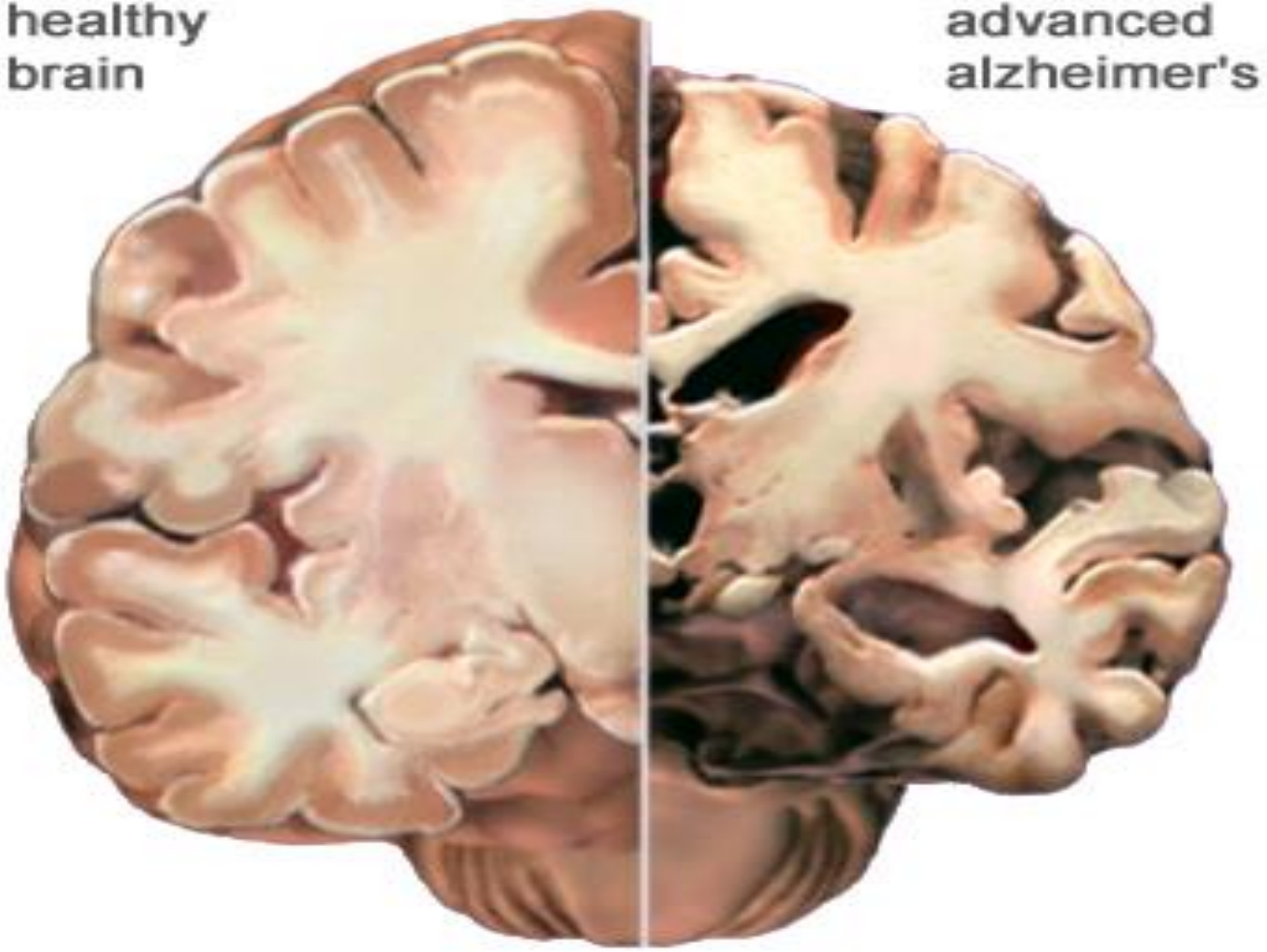
  unable to walk, get up

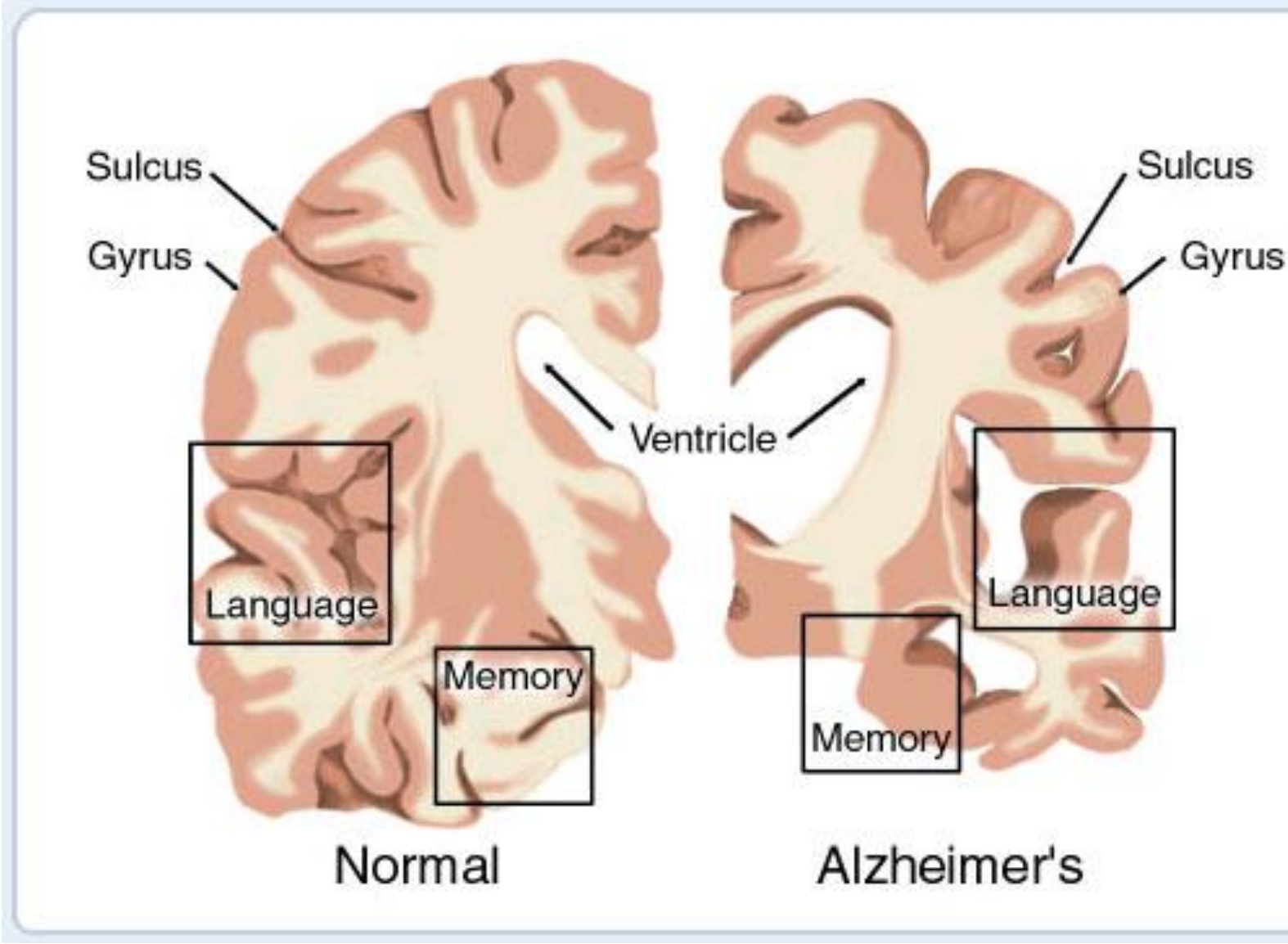
# The Brain and Alzheimer's Disease

Two major structural changes:

1. **Neurofibrillary tangles**
   - Bundles of twisted threads that are the product of collapsed neural structures (contain abnormal forms of tau protein
2. **Amyloid plaques**
   - Dense deposits of deteriorated amyloid protein, surrounded by clumps of dead nerve and glial cells


tangle
plaques
Alzheimer cells
healthy cells


healthy
brain
advanced
alzheimer's


Sulcus
Gyrus
Ventricle
Language
Memory
Normal
Sulcus
Gyrus
Language
Memory
Alzheimer's

# Causes?

Several competing hypotheses:

**Cholinergic hypothesis**

-Caused by reduced synthesis of acetylcholine

**Amyloid hypothesis**

-Abnormal breakdown; buildup of amyloid beta deposits

-Damaged amyloid proteins build to toxic levels, causing call damage and death

**Tau hypothesis**

-Caused by tau protein abnormalities

-Formation of neurofibrillary tangles

# Risk Factors

- Obesity
- High blood pressure
- Head trauma
- High cholesterol
- Being American!
  - Higher rates in
    - Japanese-Americans than Japanese
    - African-Americans than Africans
- Depression

# Medications

- **Cholinesterase inhibitors increase the levels of acetylcholine in the brain, which plays a key role in memory and learning.**

- **Aricept**
  Donepezi

  **Used to delay or slow the symptoms of AD**

  - Used for mild, moderate and severe AD
  - Does not prevent or cure AD

- **Celexa**
  Citalopram

  **Used to reduce depression and anxiety**

  - Sometimes used to help people get to sleep

- **Depakote**
  Sodium

  **Used to treat severe aggression**

  - Also used to treat depression and anxiety

- **Zoloft**
  Sertraline

  **Used to reduce depression and anxiety**

  - Sometimes used to help people get to sleep

# Transfusion Therapy and Blood and Marrow Stem Cell Transplantation, MS, MG, Paget's Dz

Dr Mazen AbuQamar

# BACKGROUND

- Blood product procurement, storage, preparation, and testing are regulated by the Food and Drug Administration (FDA)

- Blood is checked for compatibility with the recipient's Blood (cross-matched) before it is transfused, regardless of Blood type.

- Blood plasma compatibility is the inverse of red blood cell compatibility

- Alloimmunization: is an immune response to foreign antigens after exposure to genetically different cells or tissues.

# Blood Compatibility

***Antigens***

1. The surface membrane of RBC is characterized by glycoproteins (antigens).

2. More than 400 different antigens have been identified on the RBC membrane.

3. Only two antigenic systems (ABO and Rh) required before the transfusion.

4. The ABO blood group system is clinically the most significant because A and B

antigens elicit the strongest immune response.

Antibodies (or immunoglobulins) are proteins produced by B lymphocytes

The five classes of immunoglobulins are determined by differences in their heavy

# Blood Transfusion Options

***Autologous Transfusion***

1. Patient donate blood to be set aside for later transfusion the safest transfusion choice.
2. Autologous blood products must be clearly labeled and identified.
3. Autologous transfusion eliminates the risks of alloimmunization, immune-mediated transfusion reactions, and transmission disease

***Homologous Transfusion***

1. With this most common option, volunteer donors' blood products to patients randomly.
2. Before donation, volunteer donors receive information about the process, potential adverse effects, tests that will be performed on donated blood, postdonation instructions, and education regarding risk of HIV infection and signs and symptoms.
3. Donors are screened against eligibility criteria

***Directed Transfusion***

1. In directed transfusion, blood products are donated by a person for transfusion to a specified recipient.



# General Blood Donor Eligibility Criteria

- **Age:** 17 years or more
- **Weight:** Minimum 110 lb (49.9 kg)
- **Vital signs:** Afebrile, normotensive, pulse 50–100, blood pressure 180/100 g
- **Hemoglobin:** higher 12.5 g/dL

- **History:** travel to malarial areas, living in areas exposed to bovine spongiform encephalopathy, blood transfusion or tattoo within 12 months, recent surgery or pregnancy, corneal transplant, history of hepatitis or unexplained jaundice, history of blood cancer or recent cancer, history of behaviors at high risk for HIV

- **Immunizations:** Recent attenuated and live vaccines
- **Illnesses:** A variety of current illnesses may defer or disallow blood donation. Examples: clotting disorders, sickle cell disease, systemic lupus erythematosus, multiple sclerosis, Lyme disease, tuberculosis.
- **Medications:** Blood thinners, such as heparin and warfarin (Coumadin), disallow donation. Some other medications may result in deferral.



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# Blood Product Screening

***Serologic Testing***

1. Routine laboratory testing is performed to assess the compatibility.

a. ABO group and Rh type: determines the presence of A, B, and D antigens on the surface of the patient's RBCs.

b. Direct Coombs' test: determines the antibody attached to the patient's RBCs.

c. Crossmatch (compatibility test): detects agglutination of donor RBCs caused by antibodies in the patient's serum.

***Screening for Infectious Diseases***

a. Hepatitis b. HIV c. Cytomegalovirus (CMV) d. Syphilis e. Bacteria contamination that may be transmitted via blood transfusions include West Nile virus, human T-cell lymphoma virus (HTLV) 1 and 2, human herpes virus-8 (implicated as the causative agent of Kaposi's sarcoma), malaria, babesiosis (parasites that infect red blood cells and are spread by certain ticks), Chagas' disease, and yersinia (infection caused most often by eating raw or undercooked pork)



# ADMINISTRATION OF WHOLE BLOOD AND BLOOD COMPONENTS

Whole blood and blood components are administered 1- to increase the amount of oxygen being delivered to the tissues and organs, 2- to prevent or stop bleeding 3- to combat infection caused by decreased or defective WBCs or antibodies

Whole blood consists of RBCs, plasma, plasma proteins, and approximately 60 mL anticoagulant/preservative solution in a total volume of approximately 500 mL.

Massive blood loss of greater than 1,000 mL, requiring the oxygen-carrying properties of RBCs and the volume expansion provided by plasma.

An blood warmer may be indicated to prevent hypothermia and cardiac arrhythmias

Obtain blood product from blood bank. Inspect for abnormal color, cloudiness, clots, and excess air. Read instructions on the product label regarding storage and infusion. Check expiration date.

**Verify patient identification.**

a. Ask the patient to state his full name and date of birth or other identifier and compare with information on the wristband. If the patient is unable to provide two identifiers, verify identity with an individual familiar with the patient.

b. Compare the name, date of birth, and ID number on the wristband with the bag tag, transfusion form, and medical order.

c. Confirm ABO and Rh compatibility by comparing the bag label, bag tag, medical record, and transfusion form.

d. Check bag labels for expiration date and satisfactory serologic testing.



## Performance phase

1. Start infusion slowly (ie, 2 mL/minute). Remain at bedside 15-30 minutes. If there are no signs of adverse effect, increase flow to the prescribed rate.

2. Observe the patient closely and check vital signs at least hourly until 1 hour after transfusion. Report signs of adverse effect to health care provider immediately.

Acute reactions may occur at any time during the transfusion.

Observe closely for the most common acute complication associated with whole blood transfusion—circulatory overload (rise in venous pressure, distended neck veins, dyspnea, cough, crackles at bases of lungs).



# TRANSFUSION REACTIONS

Reactions can be placed into two general categories: acute and delayed.

**Acute Reactions**

1. Acute reactions may occur during the infusion or within minutes to hours after the blood product has been infused.

2. Acute reactions include allergic, febrile, septic, and hemolytic reactions, air embolism, and circulatory overload.

Patients who also receive multiple blood products within a short time frame may also be at risk for hyperkalemia, hypocalcemia, and hypothermia.

4. When a reaction is suspected, the health care provider should be notified immediately and blood bags with tubing from all products recently transfused should be returned to the blood bank for evaluation.



5. The following samples should also be obtained if an acute reaction is suspected.

a. A clotted blood specimen to examine serum for hgb and confirm RBC group and type.

b. An anticoagulated blood sample for a direct Coombs' test to determine the presence of antibody on the RBCs.

c. The first voided urine specimen to test for hemoglobinuria

6. Precautions must be taken to avoid the hemolysis of RBCs during venipuncture and sample collection because this could lead to invalid test results. Whenever possible, blood samples should be drawn from a fresh venipuncture and not from existing needles or catheters.

7. If the only symptoms are those resulting from a mild allergic reaction (eg, urticaria), extensive evaluation may not be necessary.

S&S: Lower back pain, Feeling of head, fullness; flushing, Tachycardia, tachypnea, Hypotension, vascular collapse, Hemoglobinuria, hemoglobinemia, Bleeding, Acute renal failure



**Delayed Reactions**

1. Delayed reactions occur days to years after the transfusion.

2. Delayed reactions include delayed hemolytic reactions, iron overload (hemosiderosis), GVHD, infectious diseases (eg, hepatitis B, hepatitis C, CMV, Epstein-Barr virus, malaria, HIV, HTLV).

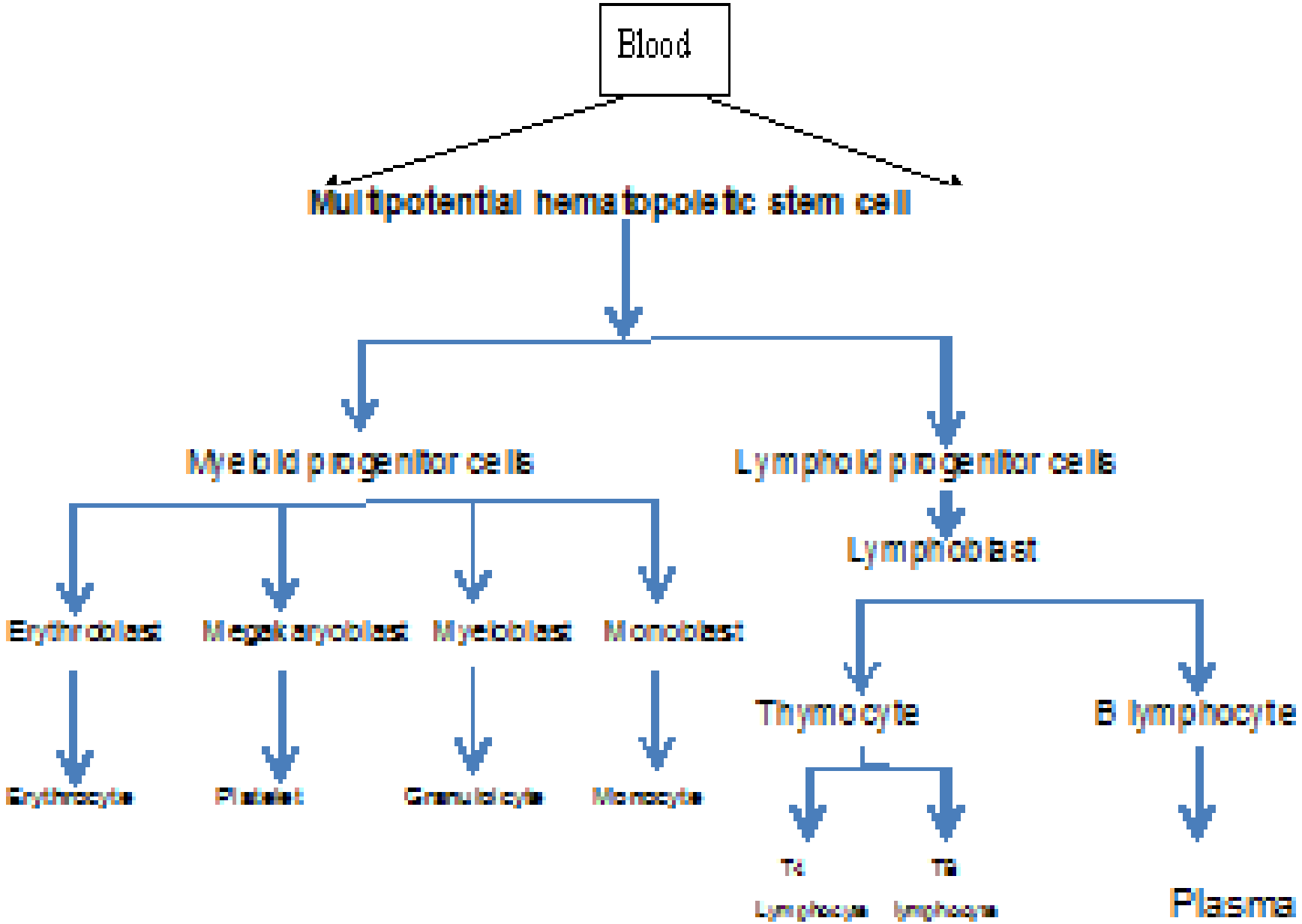
3. Symptoms of a delayed reaction can vary from mild to severe. Diagnosis may be complicated by the long incubation period between transfusion and reaction and the complexity of diagnostic tests.

4. clinical manifestation are Fever,  Mild jaundice, Decreased hematocrit



# The Hematologic System




Blood
Multipotential hematopoietic stem cell
Myeloid progenitor cells
Lymphoid progenitor cells
Lymphoblast
Erythroblast
Megakaryoblast
Myeloblast
Monoblast
Thymocyte
B lymphocyte
Plasma

# Blood

- Represents its own organ system
- Has specific functions
  - Clotting
  - Delivery of oxygen and removal of $CO_2$
  - Removal and delivery of waste products to organs that remove them

*continued*

# Blood

- Made up of solid components
  - Red blood cells
  - White blood cells
  - Platelets
  - Plasma



# Anemia

- Lower-than-normal amount of red blood cells
- Acute anemia
  - Sudden blood loss
- Chronic
  - Excessive menstrual periods
  - Slow gastrointestinal bleeding
  - Diseases affecting bone marrow



# Sickle Cell Anemia

- Genetic disease affecting RBCs
- More prevalent in certain ethnicities
  - African Americans
  - Indian or Middle Eastern descent
- Defective shape resembles a sickle
- Cells have a short life span leading to anemia



# Complications of Sickle Cell Anemia

- Destruction of spleen
- Sickle pain crisis
- Acute chest syndrome
- Priapism
- Stroke



# Treatment of Sickle Cell Anemia

- High flow supplemental oxygen
- Monitor for respiratory distress
- Monitor for signs of hypoperfusion
- Transport to stroke center if stroke is suspected



# Iron Deficiency Anemia



Iron Deficiency Anemia

- **Scope of the problem**
  - A common false assumption is that iron deficiency anemia (IDA) due to inadequate nutrition is confined to developing or underdeveloped countries.
  - Worldwide, more than 40% of children have an IDA that is frequently associated with infections.

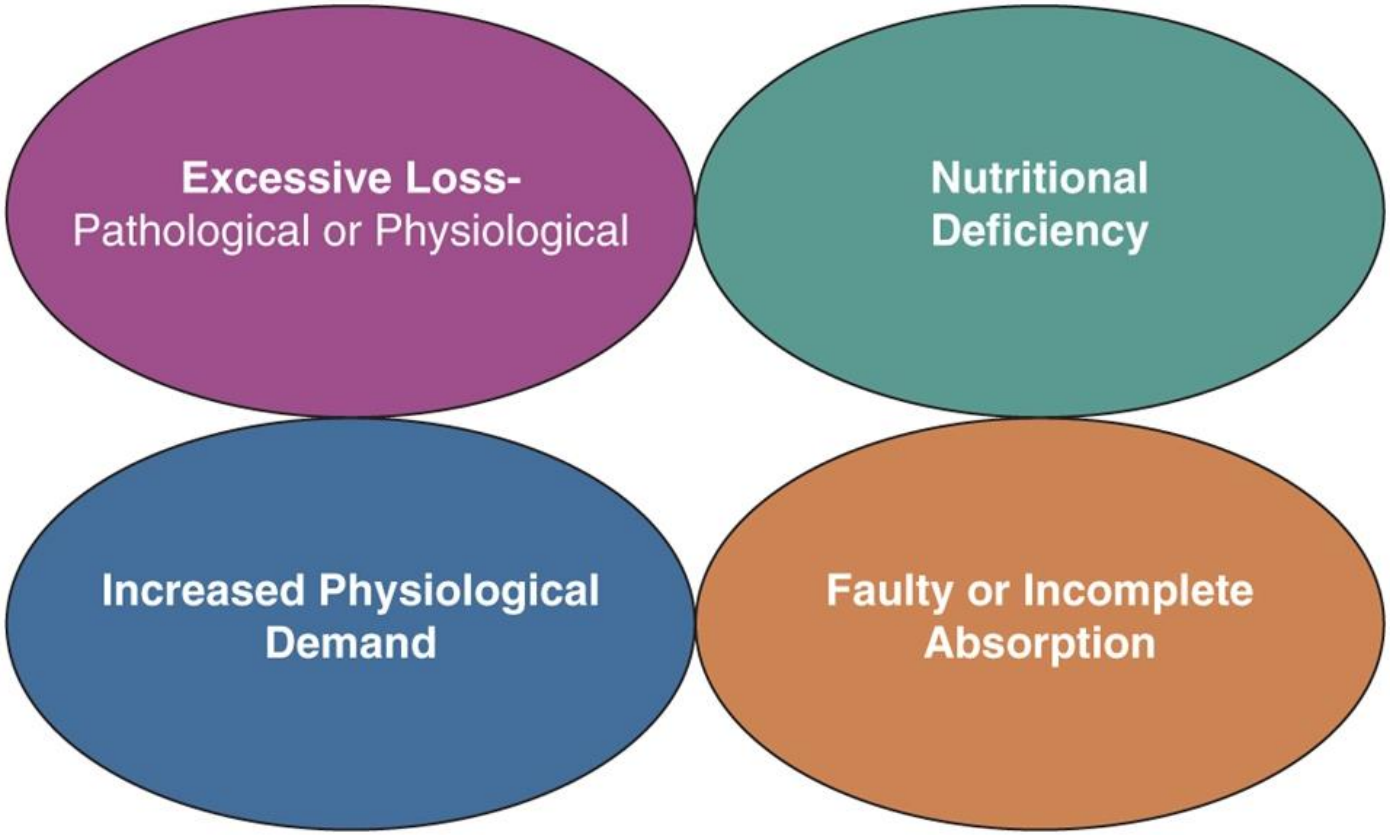


# Iron Deficiency Anemia

- Early diagnosis of iron deficiency is essential in nonanemic infants and toddlers (under 2 years of age).
- Early diagnosis of iron deficiency is equally important in pregnant women to reduce maternal–fetal morbidity.



# Factors in Iron Deficiency Anemia




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# Iron Deficiency Anemia (cont.)

**Etiology**

1. Decreased iron intake, for example, meat-poor diets
2. Faulty or incomplete iron absorption, for example, resection of small bowel, celiac disease
3. Increased iron utilization, for example, growth spurt
4. Iron loss (pathological), for example, GI bleeding, malignancy
5. Iron loss (physiological), for example, pregnancy, menstruation



# Iron Deficiency Anemia

- Humans have 35 to 50 mg of iron per kilogram of body weight.
- The average adult has 3.5 to 5.0 g of total iron.
- Iron deficiency continues to be common in Toddlers, Adolescent girls, Women of childbearing age



- **Physiology**
  - Operational iron consists of iron used for oxygen binding and biochemical reactions.
  - In humans, most operational iron is found in the heme portion of hemoglobin or myoglobin.
  - Most operational iron is incorporated into the hemoglobin molecules of erythrocytes and is recycled.
  - In normal adults, hemoglobin contains two thirds of the iron present in the body.
  - Normal iron loss is very small, amounting to less than 1 mg/day.
  - Iron is lost from the body through exfoliation of intestinal epithelial and skin cells, the bile, and through urinary excretion.
  - To compensate for this loss, the adult male has a replacement iron need of 1 mg/day.



# Iron Deficiency Anemia

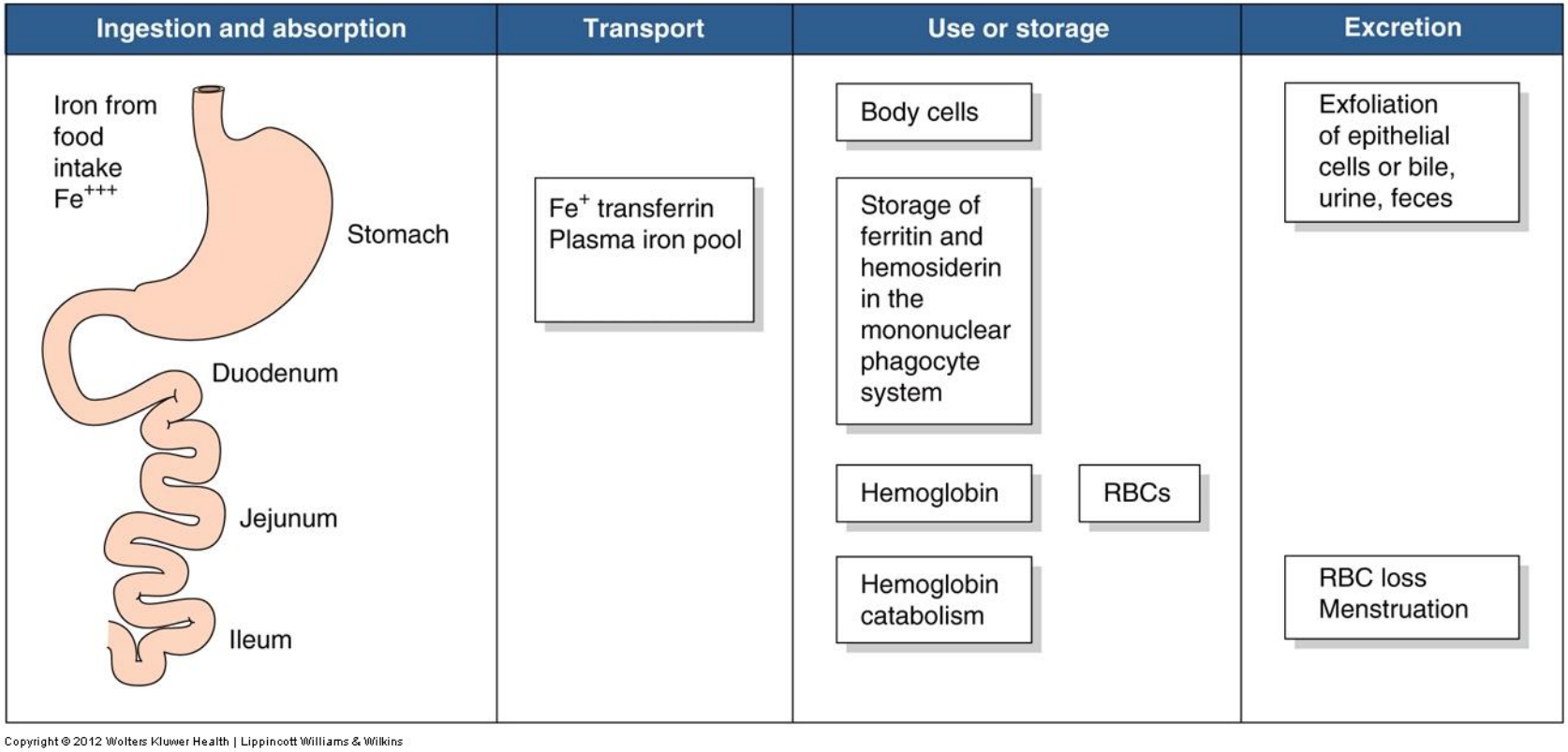



Figure 10.2 Iron physiology.

- **Iron needs in infants and children**
  - In the normal infant at term, iron stores are adequate to maintain iron sufficiency for approximately 4 months of postnatal growth.

- Breast milk and cow's milk both contain about 0.5 to 1.0 mg of iron per liter, but its bioavailability differs significantly.

- The absorption of iron from breast milk is uniquely high, about 50% on average, and tends to compensate for its low concentration.

- By contrast, only about 10% of iron in whole cow's milk is absorbed.



- Sequential phases of iron deficiency:
  - Stage 1 (Prelatent)
    - Decrease in storage iron
  - Stage 2 (Latent)
    - Decrease in iron for erythropoiesis
  - Stage 3 (Anemia)
    - Decrease in circulating red blood cell parameters &
    - Decrease in oxygen delivery to peripheral tissues



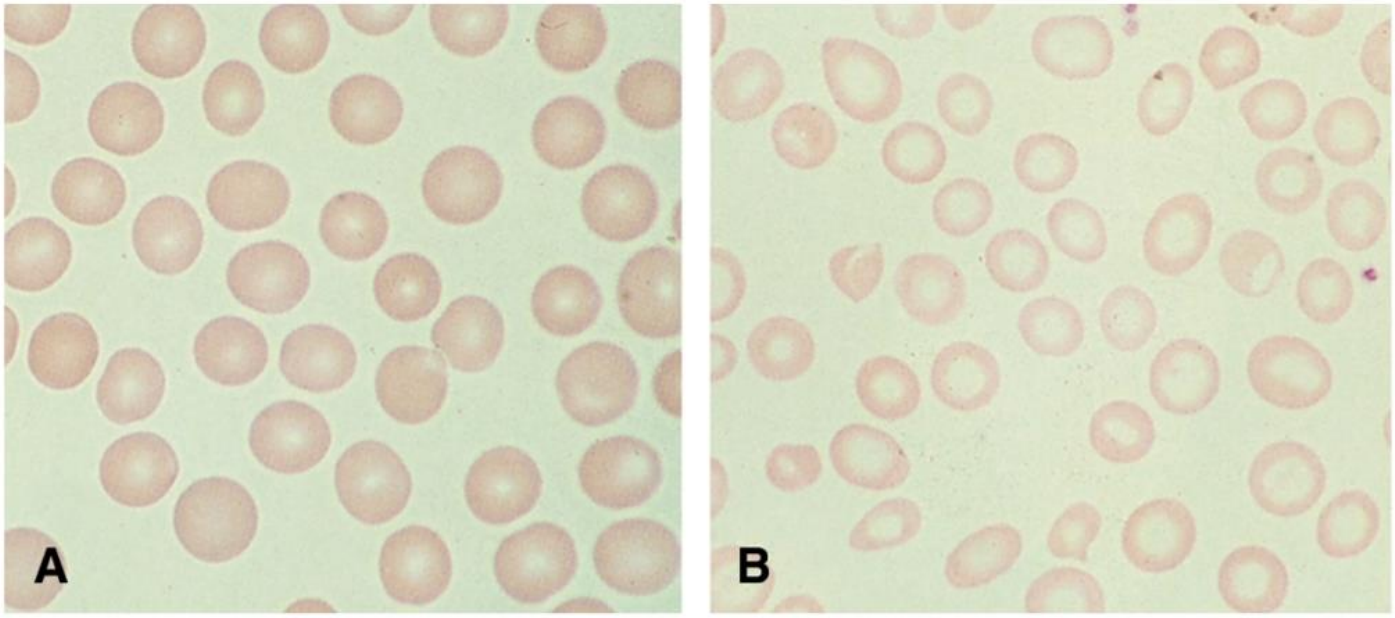
- **Clinical signs and symptoms**
  - The history and physical presentation are the typical initial observations in the diagnostic workup of a patient with symptoms of paleness, fatigue, and/or weakness.
  - Iron deficiency anemia in children is associated with psychomotor and mental impairment in the first 2 years of life.
  - Currently, more than one third of children in the United States demonstrate evidence of iron insufficiency,
    - 7% have iron deficiency without anemia, and
    - 10% have iron deficiency anemia.



- **Laboratory characteristics—hematology studies**
  - Complete blood count including observation of the peripheral blood smear and a platelet count
  - Platelet count and white blood cell count should be noted.
  - Essential red blood cell parameters:
    - Hemoglobin
    - Microhematocrit
    - Red blood cell count
    - Calculation of the red blood cell indices (MCV=mean corpuscular volume, MCH=mean corpuscular hemoglobin, and MCHC=mean corpuscular hemoglobin concentration)
    - **Average red blood cell size (MCV)**
    - **Hemoglobin amount per red blood cell (MCH)**
    - **The amount of hemoglobin relative to the size of the cell (hemoglobin concentration) per red blood cell (MCHC)**



# Iron Deficiency Anemia (cont.)




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Figure 10.5 A blood smear showing normal erythrocytes (**A**) compared with a blood smear revealing microcytic, hypochromic erythrocytes in a patient with iron deficiency anemia (**B**). (Reprinted with permission from Willis MC. Medical Terminology: A Programmed Learning Approach to the Language of Health Care. Baltimore, MD: Lippincott Williams & Wilkins, 2002.)

# Iron Deficiency Anemia (cont.)

**TABLE 10.3 Red Blood Cell Characteristics in Children and Adults**

| Age | Lowest Normal Hemoglobin (g/dL) | Normal RBC MCV | Fetal Hemoglobin (%) |
|---|---|---|---|
| Birth | 14.0 | 100–130 | 55–90 |
| 3–6 months | 10.5 | 75–90 | 5–25 |
| 1–4 years | 11.0 | 70–85 | <2 |
| 4 years to puberty | 11.5 | 75–90 | <2 |
| Adult (female) | 12.0 | 80–95 | <2 |
| Adult (male) | 14.0 | 80–95 | <2 |

*Source*: Greer JP, et al. *Wintrobe's Clinical Hematology*, 11th ed., [(Table 27.1 Red Blood Cell Characteristics in Childhood]), 2004:948.

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# Anemia of Inflammation (AOI) or Anemia of Chronic Disorders (ACD)

- **Etiology**
  - Anemia of chronic diseases or disorder (ACD) or anemia of inflammation is another common form of anemia.
  - Anemia results in illnesses as diverse as inflammation, infection, malignancy, or various systemic diseases.
  - Approximately half of AOI/ACD cases are caused by subacute or chronic infections.



## Microbial agents associated with anemia of inflammation

- Bacterial, for example, *M. tuberculosis*
- Fungal, for example, *C. neoformans*
- Viral, for example, HIV, Cytomegalovirus

Other cases may be caused by the following:

Neoplasms

- Rheumatoid arthritis
- Rheumatic fever
- Systemic lupus erythematosus (SLE)
- Uremia
- Chronic liver disease



# Pathophysiology

- AOI/ACD is a hypoproliferative defect not related to any nutritional deficiency.
- The principal pathogenesis of ACD is believed to be related to ***hepcidin***, a small plasma protein, that is a key molecule in controlling iron absorption and recycling.

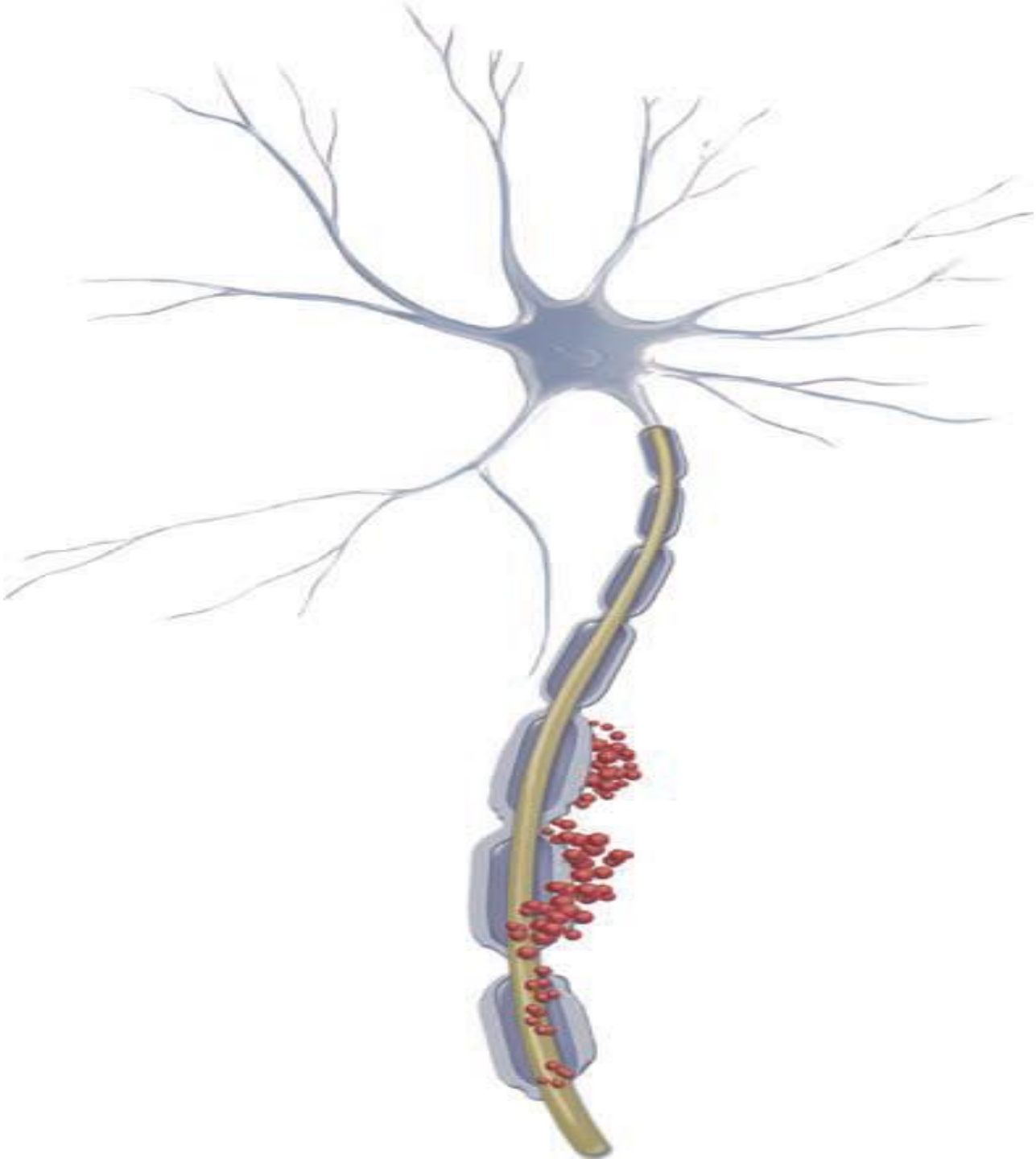


Characteristics of AOI associated with malignancy can be as follows:

- Decreased erythrocyte production because of direct bone marrow infiltration by malignant tumor cells or by primary marrow cell malignancies
- Production and release of TNF-α and IL-1 by macrophages
- Increased erythrocyte destruction present in immune or microangiopathic hemolytic anemia
- Acute and chronic blood loss
- Toxic effects of invasive therapy (e.g., chemotherapy or radiation therapy)
- Indirect multiple causes such as anemia associated with malignant disease, anemia associated with major organ failure, and various hemolytic anemias

- Treatment of the underlying cause of anemia is the most direct approach.





# MULTIPLE SCLEROSIS

- MS is a chronic, frequently progressive neurologic disease of the CNS of unknown etiology and uncertain trajectory.
- MS is the occurrence of small patches of demyelination of the white matter of the optic nerve, brain, and spinal cord.
- MS is distinguished by exacerbations and remissions of symptoms over the course of the illness.
- MS is the most common CNS disease among young adults and the third leading cause of disability in the United States.
- It is estimated that 400,000 Americans have this disorder of the brain and spinal cord, causing disruption of electrical messages from the brain to the peripheral nervous system.
- Most people with MS are diagnosed between the ages of 20 and 50, with at least two to three times more women than men and white

**Pathophysiology and Etiology**

1. *Demyelination* refers to the destruction of the myelin, the fatty and protein material that covers nerve fibers in the brain and spinal cord .

2. Demyelination results in disordered transmission of nerve impulses.

3. Inflammatory changes lead to scarring of the affected nerve fibers.

4. Cause is unknown but may possibly be related to autoimmune dysfunction, genetic susceptibility, or an infectious process.

**Clinical Manifestations**

- Lesions can occur anywhere within the white matter of the CNS. Symptoms reflect the location of the area of demyelination.

1. Fatigue and weakness.
2. Abnormal reflexes—absent or exaggerated.
3. Vision disturbances—impaired and double vision, nystagmus.
4. Motor dysfunction—weakness, tremor, incoordination.
5. Sensory disturbances—paresthesias, impaired deep sensation.
6. Impaired speech—Dysarthria (weak muscles speech is characterized by slurred or slow speech)
7. Urinary dysfunction—hesitancy, frequency, urgency, retention, incontinence; upper UTI. Urinary dysfunction.
8. Neurobehavioral syndromes—depression, cognitive impairment, emotional lability.
9. Symptoms of MS are often unpredictable, varying from person to person and from time to time in the same person.

# Diagnostic Evaluation

1. Establishing a definitive diagnosis is often difficult, with much uncertainty concerning prognosis once the diagnosis is made.

2. Serial brain MRI studies have proved to be useful for diagnosing and monitoring patients with MS—show small plaques scattered throughout white matter of CNS.

3. Electrophoresis study of CSF shows abnormal IgG antibody.

4. Visual, auditory, and somatosensory evoked potentials—slowed conduction is evidence of demyelination.

# Management

- MS treatment is direct treatment of MS and symptoms from MS.
- Treatment is aimed at relieving symptoms & helping the patient function.
- Corticosteroids are used to decrease inflammation, shorten duration of relapse or exacerbation.
- Immunosuppressive agents may stabilize the course.
- Interferon (Rebif, Avonex, Betaseron) are being used for treatment of rapidly progressing symptoms in some patients.
- Copolymer-1, a mixture of synthetic polypeptides composed of four amino acids, has been effective in reducing relapse rates and disability.
- A chemotherapeutic agent used to reduce neurologic disability and frequency of clinical relapses.

# Management

- Plasmapheresis (plasma exchange) is only considered for the 10% who do not respond well to standard corticosteroid treatment.
- Treatment of spasticity with such agents as baclofen
- Treatment of depression with antidepressant drugs
- Bladder management with anticholinergics, catheterization for drainage, prophylactic antibiotics
- Bowel management with stool softeners, bulk laxative, suppositories
- Physical therapy, occupational therapy
- Control of dystonia and pain with carbamazepine (Tegretol)

# Complications

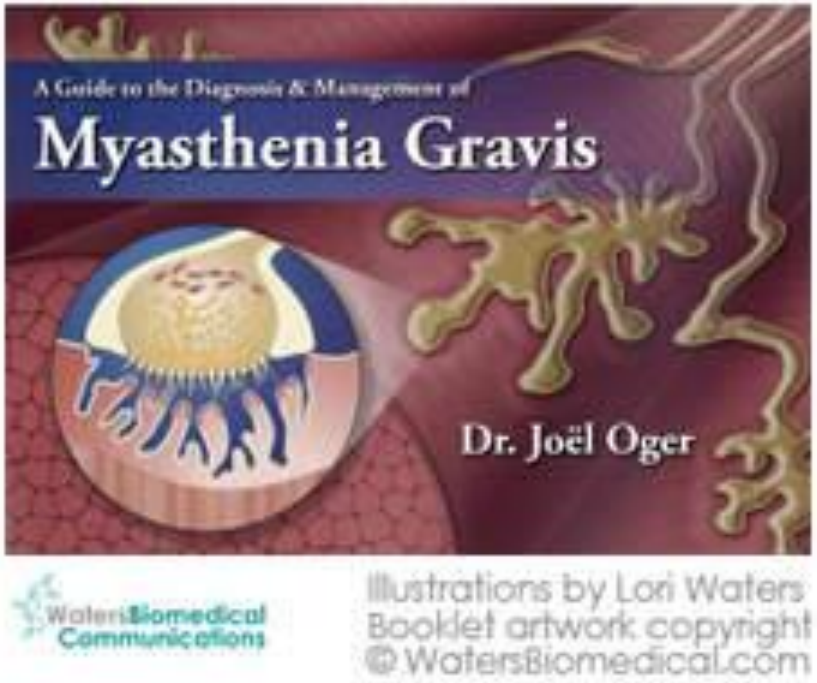
1. Respiratory dysfunction

2. Infections: bladder, respiratory, sepsis

3. Complications from immobility

4. Speech disorders such as dysarthria

# MYASTHENIA GRAVIS



# Myasthenia gravis (MG)

- is a chronic autoimmune disorder affecting the neuromuscular transmission of impulses in the voluntary muscles of the body.
- MG is characterized by fluctuating weakness increased by exertion.
- Weakness increases during the day and improves with rest.
- Presentation and progression vary due to an antibody-mediated attack against acetylcholine receptors at the neuromuscular junction.
- Loss of acetylcholine receptors leads to a defect in neuromuscular transmission.
- Cardinal features are muscle weakness and fatigue.

# Myasthenia gravis (MG)

- About 80% to 90% of patients with MG have serum antibodies to acetylcholine.
- Thyroid gland abnormalities are present in 80% of patients.
- Thymic tumor is the most important known cause; 10% of patients with MG have a thymoma.
- *Cholinergic crisis* can result from overmedication with anticholinergic drugs, which release too much acetylcholine at the neuromuscular junction.
- *Brittle crisis* occurs when the receptors at the neuromuscular junction become insensitive to anticholinesterase medication.
- Women are three times more susceptible to developing MG than men.

# Clinical Manifestations

- Extreme muscular weakness and easy fatigability.
- Vision disturbances—diplopia and ptosis from ocular weakness.
- Facial muscle weakness causes a masklike facial expression.
- Dysarthria & dysphagia from weakness of laryngeal and pharyngeal muscles.
- Proximal limb weakness
- Respiratory muscle weakness can be life-threatening.
- Impending myasthenic crisis may be triggered by respiratory infection, aspiration, physical/emotional stress, and changes in medications.

**Symptoms include:**

a. Sudden respiratory distress.

b. Signs of dysphagia, dysarthria, ptosis, and diplopia.

c. Tachycardia, anxiety.

d. Rapidly increasing weakness of extremities and trunk.

# Diagnostic Evaluation

- Serum test for acetylcholine receptor antibodies—positive in up to 90% of pt
- Edrophonium (Tensilon) test—I.V. injection relieves symptoms temporarily. After injection, a marked but temporary improvement in muscle strength suggests MG. It also differentiates myasthenic crisis from cholinergic crisis.
- Electrophysiologic (EMG) testing—reveals decremental response to repetitive nerve stimulation.
- Thyroid function tests should be done to evaluate for coexistent thyroid disease and CT scan to assess enlargement of thymus gland.
- Chest CT scan is mandatory to identify thymoma in all cases of MG. This is especially true in older individuals.

# Management

- Oral anticholinesterase drugs, such as neostigmine & pyridostigmine (Mestinon), are first-line treatments for mild MG, enhancing neuromuscular transmission.
- Immunosuppressive drugs, such as prednisone, are used when weakness is not adequately controlled by anticholinergic medication.
- Azathioprine (Imuran) may be added as a steroid-sparing agent. Immunosuppressant treatment is often permanent.
- Plasmapheresis removes antibodies from the blood
- Thymectomy is indicated for patients with tumor or hyperplasia of the thymus gland (thymectomy is not usually beneficial in late-onset patients).

# Complications

1. Aspiration

2. Complications of decreased physical mobility

3. Respiratory failure

# PAGET'S DISEASE (OSTEITIS DEFORMANS)

- is a skeletal disorder resulting from excessive osteoclastic activity, affecting the long bones, pelvis, lumbar vertebrae, and the skull

**Pathophysiology and Etiology**

- The cause of this disease is unknown, although there is evidence of familial tendency (25% to 40% have at least one affected relative).
- More common in men than women.
- Rare before age 40 and increases as age does—12% after age 80.
- May be caused by infection from blood-borne viruses.
- After acute viremia, osteoclasts become chronically infected, stimulating osteoclastic proliferation.



# Clinical Manifestations

- Generally, asymptomatic.
- Most common symptoms are pain and predisposition to fracture.
- Pagetic lesions can lead to OA, joint destruction, spinal deformity.
- Decrease in hearing as a result of skull enlargement.
- Tinnitus or vertigo may also occur.
- Rarely, heart failure, hypertension, atherosclerosis, and aortic valve calcification occur.
- Malignant bone tumor.



# Diagnostic Evaluation

- Elevated serum alkaline phosphatase.
- Serum calcium, phosphorus, and albumin levels usually normal.
- Generally confirmed with radiologic examinations.
- Bone scans can evaluate pagetic activity.
- Bone biopsy.

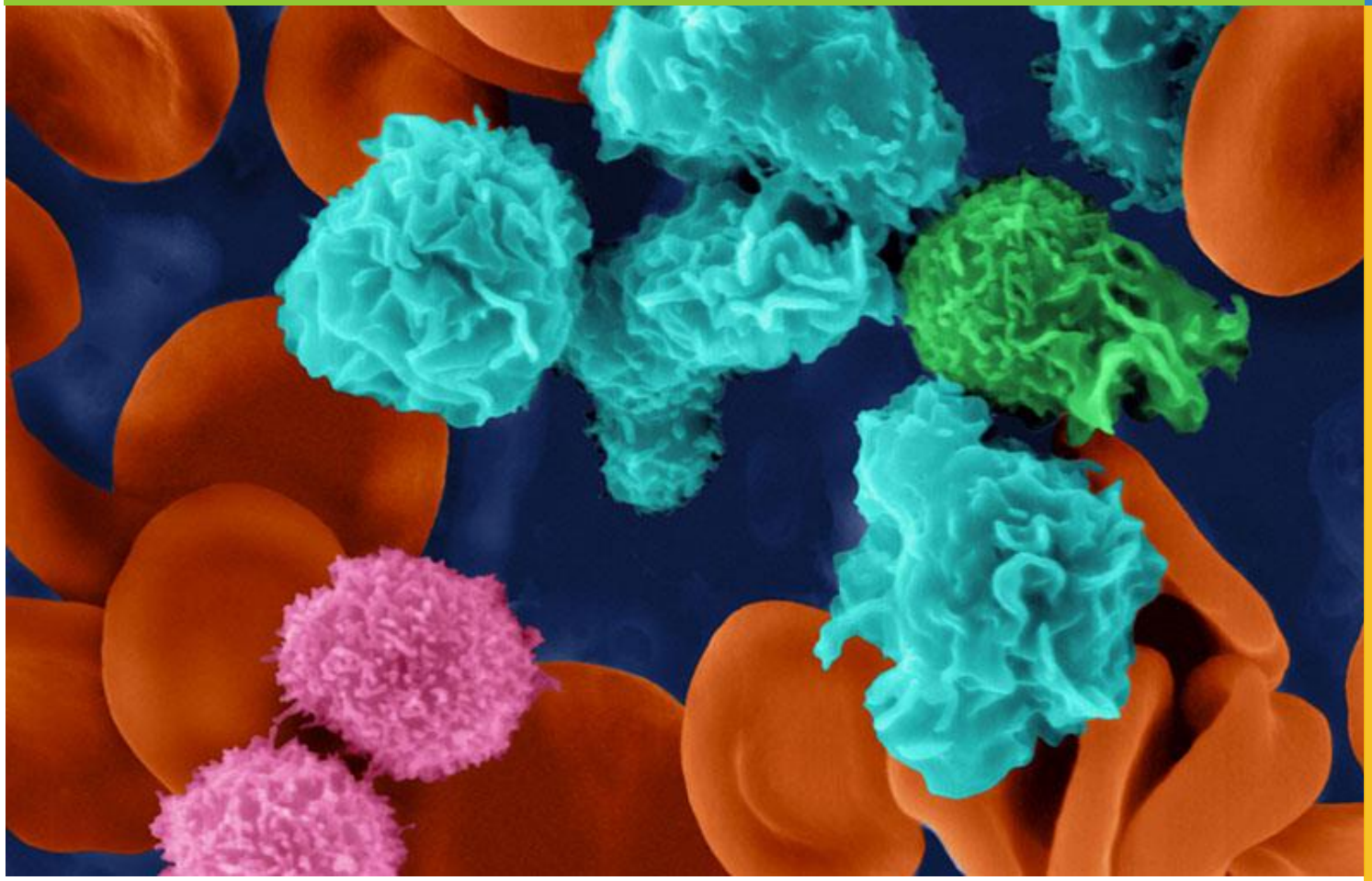


# Management

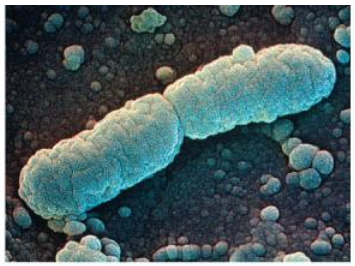
- No treatment for asymptomatic Paget's.
- Pain management—NSAIDs, aspirin.
- Medications—calcitonin is the main medication used for this disease.
- Other medications used to block bone resorption—the bisphosphonates etidronate disodium (Didronel); an antineoplastic agent, plicamycin .
- Tibial osteotomy done to realign knees and relieve pain.



# Body Defenses



# Defense Against Disease



**Nonspecific External Barriers**
skin, mucous membranes

If these barriers are penetrated,
the body responds with

**Innate Immune Response**
phagocytic and natural killer cells,
inflammation, fever

If the innate immune response is insufficient,
the body responds with

**Adaptive Immune Response**
cell-mediated immunity, humoral immunity

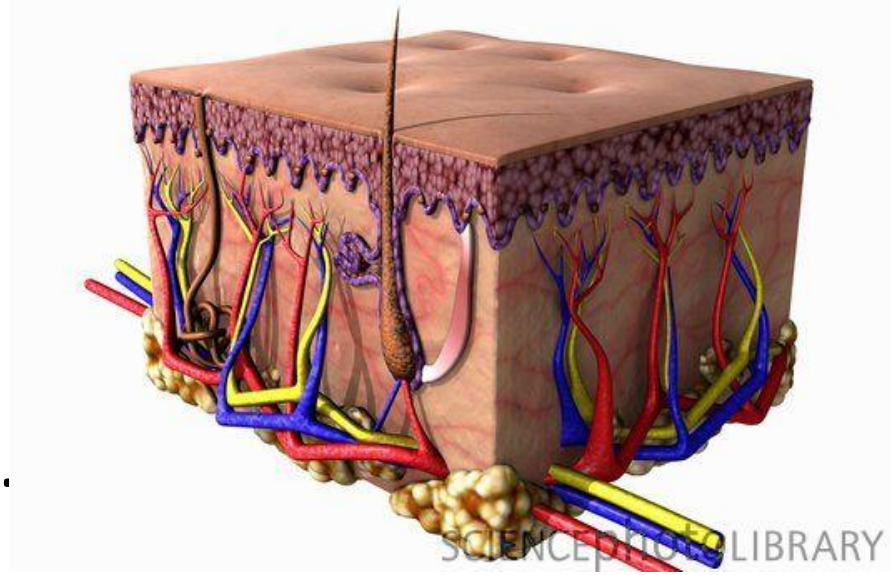
# First line of defense

- Non-specific defenses are designed to prevent infections by viruses and bacteria. These include:
  - Intact skin
  - Mucus and Cilia
  - Phagocytes



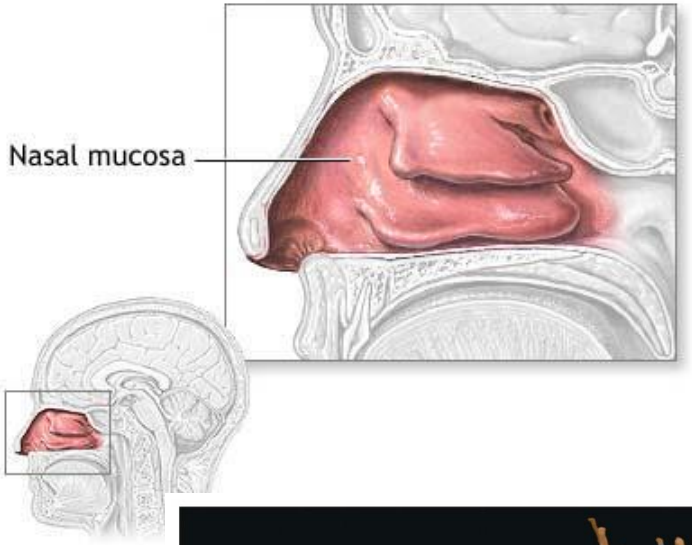
# Role of skin

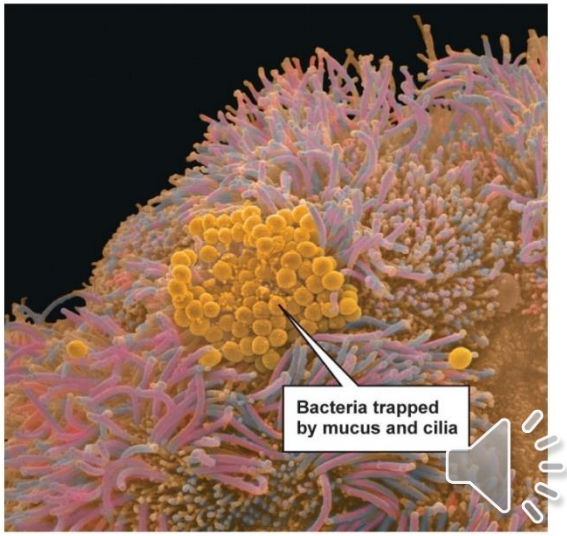
- Dead skin cells are constantly sloughed off, making it hard for invading bacteria to colonize.
- Sweat and oils contain anti-microbial chemicals, including some antibiotics.



# Role of mucus and cilia

- Mucus contains lysozymes, enzymes that destroy bacterial cell walls.
- The normal flow of mucus washes bacteria and viruses off of mucus membranes.
- Cilia in the respiratory tract move mucus out of the lungs to keep bacteria and viruses out.

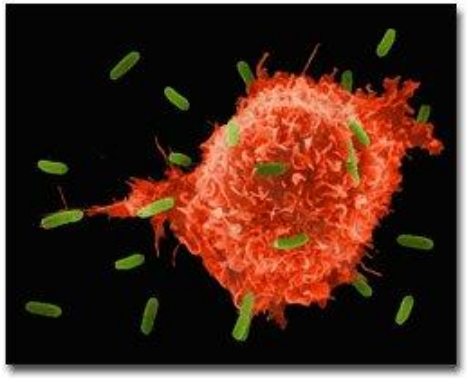


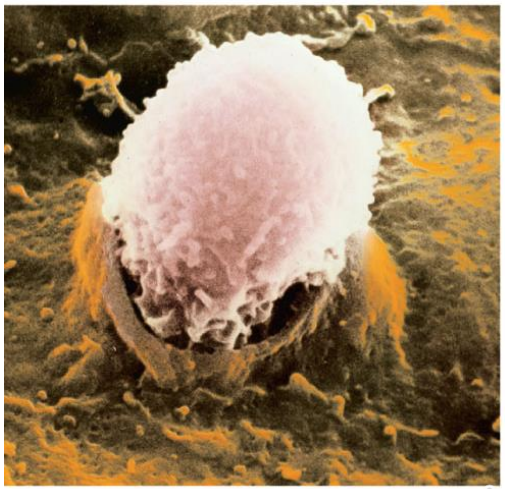




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# Role of phagocytes

- Phagocytes are several types of white blood cells (including macrophages and neutrophils) that seek and destroy invaders. Some also destroy damaged body cells.
- Phagocytes are attracted by an inflammatory response of damaged cells.





# Role of inflammation

- Inflammation is signaled by mast cells, which release histamine.
- Histamine causes fluids to collect around an injury to dilute toxins. This causes swelling.
- The temperature of the tissues may rise, which can kill temperature-sensitive microbes.

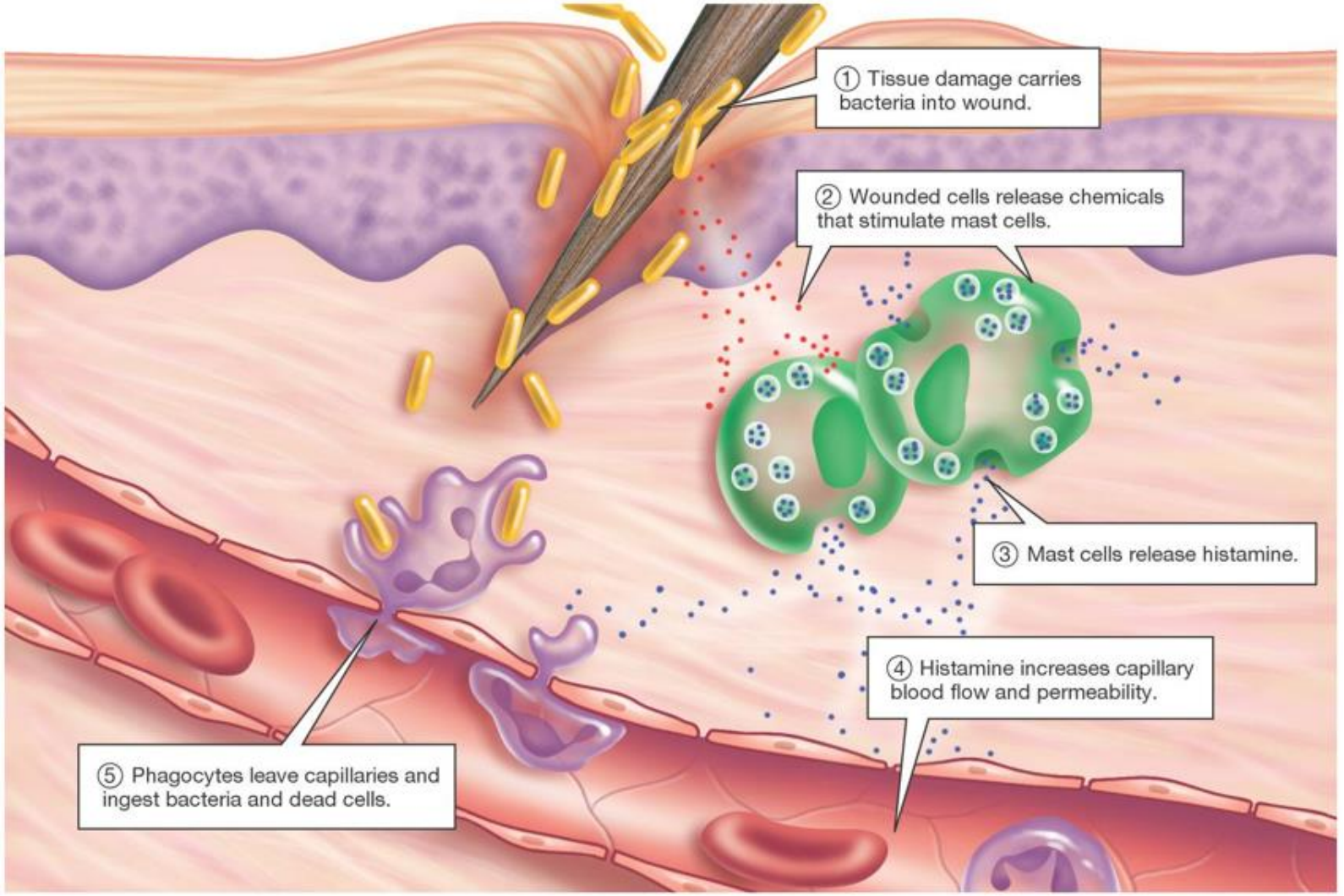


# Role of fever

- Fever is a defense mechanism that can destroy many types of microbes.
- Fever also helps fight viral infections by increasing interferon production.
- While high fevers can be dangerous, some doctors recommend letting low fevers run their course without taking aspirin or ibuprofen.



# Ouch!


① Tissue damage carries bacteria into wound.
② Wounded cells release chemicals that stimulate mast cells.
③ Mast cells release histamine.
④ Histamine increases capillary blood flow and permeability.
⑤ Phagocytes leave capillaries and ingest bacteria and dead cells.

# Major players

- The major players in the immune system include:
  - Macrophage
  - T cells (helper, cytotoxic, memory)
  - B cells (plasma, memory)
  - Antibodies



# Some vocabulary:

- Antibody: a protein produced by the human immune system to tag and destroy invasive microbes.
- Antigen: any protein that our immune system uses to recognize "self" vs. "not self."

- Immunity:

  a. *Humoral*—process by which B lymphocytes produce circulating antibodies to act against antigens.

  b. *Cell-mediated*—portion of the immune system in which the participation of T lymphocytes and macrophages is predominant.



- *Mast cell*—tissue cell similar to peripheral blood basophil, which contains granules with chemical mediators.
- *Basophil*—leukocyte with large granules that contain histamine.
- *Atopy*—term that refers to the ability to produce immunoglobulin E antibodies to common allergens.
- *Hypersensitivity*—reaction to an antigen after reexposure (there are four types); type I (immediate) and type IV (delayed) are considered allergic reactions



# Helper T cells

- Helper T-cells have receptors for recognizing antigens. If they are presented with an antigen, they release cytokines to stimulate B-cell division.
- The helper T-cell is the key cell to signal an immune response. If helper T-cells are disabled, as they are in people with AIDS, the immune system will not respond.



# B cells

- B-cells in general produce antibodies. Those with antibodies that bind with the invader's antigen are stimulated to reproduce rapidly.
- B-cells differentiate into either plasma cells or memory B-cells. Plasma cells rapidly produce antibodies. Memory cells retain the "memory" of the invader and remain ready to divide rapidly if an invasion occurs again.



# “Killer” T cells

- While B-cells divide and differentiate, so do T-cells.
- Some T-cells become cytotoxic, or “killer” T-cells. These T-cells seek out and destroy any antigens in the system, and destroy microbes “tagged” by antibodies.
- Some cytotoxic T-cells can recognize and destroy cancer cells.



# Five major classes of immunoglobulins.

**IgM**—constitutes 10% of immunoglobulin pool; found mostly in intravascular fluid and primarily engaged in initial defense.

**IgG**—major immunoglobulin that accounts for 70% to 75% of secondary immune responses and combats tissue infection.

**IgA**—15% to 20% of immunoglobulins; predominantly found in seromucous secretions (such as saliva, tears) in which it provides a primary defense mechanism.

**IgD**—less than 1% of immunoglobulin pool; found on circulating B lymphocytes, but function is unknown.

**IgE**—only a trace found in serum; attaches to surface membrane of basophils and mast cells; responsible for immediate types of allergic reactions.



***Histamine***—bioactive amine stored in granules of mast cells and basophils.

***Prostaglandins***—potent vasodilators and potent bronchoconstrictors.

***Cytokines—***control and regulate immunologic functions

***Proteases***—enzymes, such as tryptase and chymase, increase vascular permeability.



- Immunotherapy (biologic therapy):

is a type of cancer treatment that boosts the body's natural defenses to fight the cancer. It uses substances made by the body or in a laboratory to improve or restore immune system function.

Immunotherapy may work in these ways:

- Stopping or slowing the growth of cancer cells
- Stopping cancer from spreading to other parts of the body
- Helping the immune system work better at destroying cancer cells

There are several types of immunotherapy, including:

- Monoclonal antibodies
- Non-specific immunotherapies
- Oncolytic virus therapy
- T-cell therapy
- Cancer vaccines

# Monoclonal antibodies

- When the body's immune system detects something harmful, it produces antibodies. Antibodies are proteins that fight infection.
- Monoclonal antibodies are a specific type of therapy made in a laboratory. They may be used in a variety of ways:
  - can be used as a **targeted therapy** to block an abnormal protein in a cancer cell.
  - can also be used as an immunotherapy. For example, some monoclonal antibodies attach to specific proteins on cancer cells. This flags the cells so the immune system can find and destroy those cells.
  - releasing the brakes on the immune system so it can destroy cancer cells. PD-1/PD-L1 and CTLA-4 pathways are critical to the immune system's ability to control cancer growth. These pathways are often called "immune checkpoints." Many cancers use these pathways to escape the immune system. The immune system responds to the cancer by blocking these pathways with specific antibodies called immune checkpoint inhibitors.

**Non-specific immunotherapies**
help the immune system destroy cancer cells. Most non-specific immunotherapies are given after or at the same time as another cancer treatment

Two common non-specific immunotherapies are:

- **Interferons.** Interferons help the immune system fight cancer and may slow the growth of cancer cells. Side effects of interferon treatment may include flu-like symptoms, an increased risk of infection, rashes, and thinning hair.
- **Interleukins.** Interleukins help the immune system produce cells that destroy cancer. It is used to treat kidney cancer and skin cancer, including melanoma. Common side effects of IL-2 treatment include weight gain and low blood pressure. Some people may also experience flu-like symptoms.

- **Oncolytic virus therapy**

Oncolytic virus therapy uses genetically modified viruses (talimogene laherparepvec (Imlygic), or T-VEC) to kill cancer cells. The virus enters the cancer cells and makes copies of itself. As a result, the cells burst and die. As the cells die, they release specific substances called antigens. This triggers the patient's immune system to target all the cancer cells in the body that have those same antigens. The virus does not enter healthy cells. (melanoma).

<u>Side effects can include:</u>

- Fatigue
- Fever
- Chills
- Nausea
- Flu-like symptoms
- Pain at the injection site

- **T-cell therapy**

T cells are immune cells that fight infection. Some T cells are removed from a patient's blood. Then, the cells are changed in a laboratory so they have specific proteins called receptors. The receptors allow those T cells to recognize the cancer cells. The changed T cells are grown in large numbers in the laboratory and returned to the patient's body. Once there, they seek out and destroy cancer cells. This type of therapy is called chimeric antigen receptor (CAR) T-cell therapy.

- **Cancer vaccines**

A vaccine is another method used to help the body fight disease. A vaccine exposes the immune system to an antigen. This triggers the immune system to recognize and destroy that antigen or related materials. There are 2 types of cancer vaccines: prevention vaccines and treatment vaccines.

# Features of Immunotherapy

1. Specific allergens are identified by skin or blood testing.
2. Serial injections are begun that contain extracts from identified allergens (allergy vaccine).
3. Initially, a small amount of dilute allergy vaccine is given, usually at weekly intervals.
4. Amount and concentration are slowly increased to maximum tolerable dose.
5. The maintenance dose is injected every 2 to 4 weeks for period of several years to achieve maximal benefit.
6. Several allergens are now standardized (dust mite, cat, grass pollens).

# Human Assist



# Helping the immune system

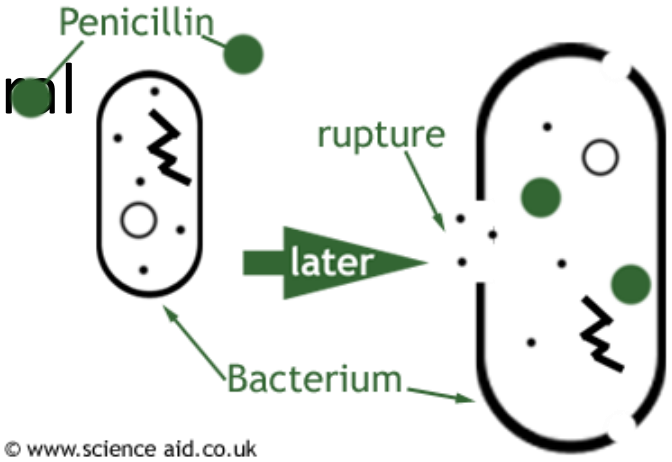
- Medical science has created to systems for augmenting the human immune system:
  - Antibiotics (NOT the same as antibodies)
  - Vaccines





# How antibiotics work

- Antibiotics help destroy bacteria (but not viruses).
- Antibiotics work in one of several ways:
  - Slowing bacteria reproduction.
  - Interfering with bacterial cell wall formation.



# Vaccine history

- Vaccination: (From *vacca*, Latin for cow.) Invented by Edward Jenner in 1796. Jenner knew that dairy maids who had contracted cowpox never got smallpox. He inoculated a boy with secretions from cowpox sores, and showed the boy was immune to smallpox.



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# Vitamin C for colds?

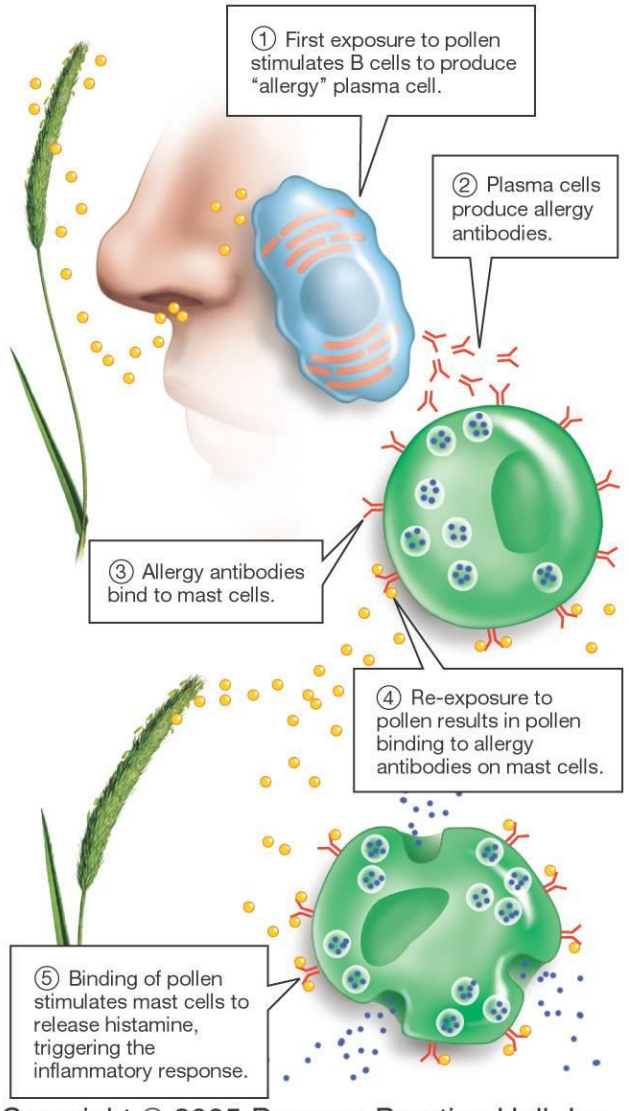
- Vitamin C is necessary for making collagen, and for many body functions.
- Absorption of Vitamin C increases during illness. It also has a very slight antihistamine effect.
- Vitamin C won't cure a cold, but may support some aspects of immune response.





# Achoo!

- Pollen is a harmless protein, yet we can become allergic to it.
- Most of the symptoms are caused by histamines released by mast cells. That is why antihistamines are used to treat allergies.



# Autoimmune disorders

- Autoimmune disorders occur when the immune system fails to recognize a protein as “self” and launches an attack.
- Multiple sclerosis, lupus, and rheumatoid arthritis are examples. None of these can be cured, but drugs can help slow the progress of these diseases.

# Cancer

- Cancer occurs when the mechanisms that control cell division fail, and body cells divide out of control.
- Cytotoxic T-cells can recognize and destroy these cells. But if division is too rapid, the T-cells cannot keep up.
- Some cancer research involves assisting cytotoxic T-cells in recognizing and destroying cancer cells.

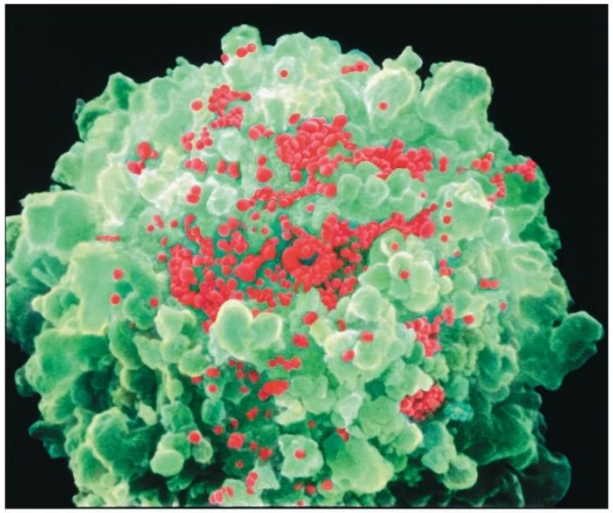
# AIDS

- AIDS (Acquired Immune Deficiency Syndrome) is caused by an infection by the HIV (Human Immunodeficiency Virus), which attacks and destroys T-helper cells. Because it attacks the immune system directly, finding a vaccine has been difficult.

- Most notably, crucial immune cells called CD4+ T cells are disabled and killed during the typical course of infection.
- A healthy, uninfected person usually has 800 to 1,200 CD4+ T cells per cubic millimeter (mm3) of blood. During HIV infection, CD4+ T cell count falls below 200/mm3,

- Some drugs can slow down HIV reproduction, but no cure exists yet. Prevention is still the best “cure.”

# AIDS

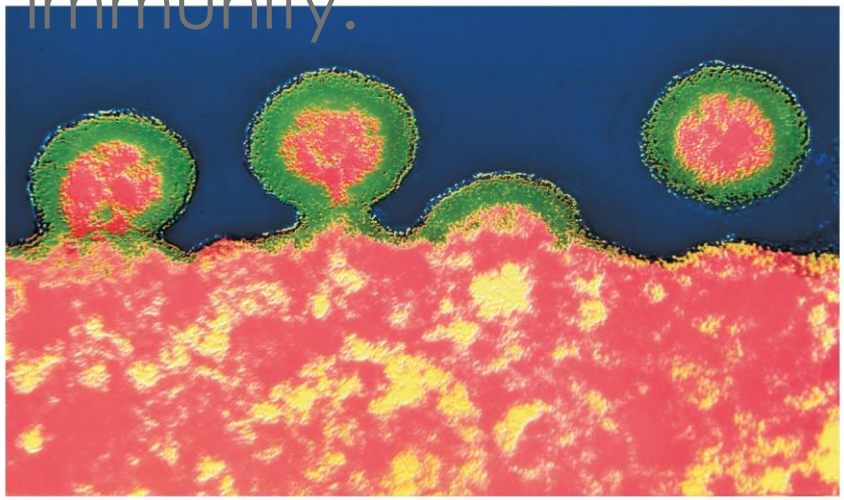


(a) HIV (red) on the surface of a helper T cell

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The HIV virus fools helper T-cells into thinking its proteins are "self," and so is able to infect the cells that trigger specific immunity.

The virus forces T-cells to make more viruses, killing the T-cells when the new viruses burst out.



(b) HIV emerging from a helper T cell

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# The infectious diseases transmitted by blood transfusion:

Introduction.

Criteria for blood donation.

How much blood can be taken during blood donation?

Factors that play a role in establishment of blood transfusion infection.

The infectious microbes that transmitted by blood transfusion:

1- HIV 2-HTLV 3-Hepatitis B and C

4-Cytomegalovirus (CMV) 5-Epstein-Barr virus.

6-Human Parvovirus (B19) 7-Human Herpesvirus 8.

8- Bacterial contamination. 9-Syphilis 10-Malaria.

# Introduction:

**Blood transfusion** is the process of receiving blood products into one's circulation intravenously.

Transfusions are used in a variety of medical conditions to replace lost components of the blood such as:

1-Inherited blood diseases: Thalassemia, Hemophilia, and Sickle cell anemia.

2- Hemolytic anemia of newborn.

3- Bleeding : Post –traumatic, or operative.

4- Malignancy: Leukemia.

5- Other conditions: hepatic coma, kidney failure.

# Criteria for blood donation:

Potential donors are evaluated for Recipient and Donor safety according to the following criteria:

1- Age: 18-55 years old.

2-Weight: 50 Kg or above.

3- Pulse Rate: Normal.

Body temperature: Normal.

Blood pressure: Normal.

Hemoglobin concentration: Normal.

4-The Medical History: Epilepsy, Psychotic disorders, abnormal bleeding tendencies, Thalassemia, Sickle cell anemia, malignancy are permanently unfit for blood donation.

5- Infection: HIV, Hepatitis, Malaria, syphilis.

# How much blood can be taken during blood donation?

The amount of blood drawn varies from 200 milliliters to 550 milliliters depending on the country, but **450-500 milliliters** is typical.

The blood is usually stored in a flexible plastic bag that also contains sodium citrate, phosphate, dextrose, and sometimes adenine.

# Factors that play a role in establishment of blood transfusion infection:

**1- Viral Window Period.**

The window period is the period between the onset of viral infection and the appearance of detectable antibodies to the virus.

Antibodies are produced from about three weeks after infection and usually become detectable by four to six weeks after infection. This four- to six-week period between infection and a positive test is called the **window period**.

**2- The Genetic Vertical transmission of viruses**.

**3- Donor immune status** (Asymptomatic immuno-competent patients).

**4- Laboratory** and **personal** error.

**5- Bacterial contamination**.

# The infectious microbes that transmitted by blood transfusion:

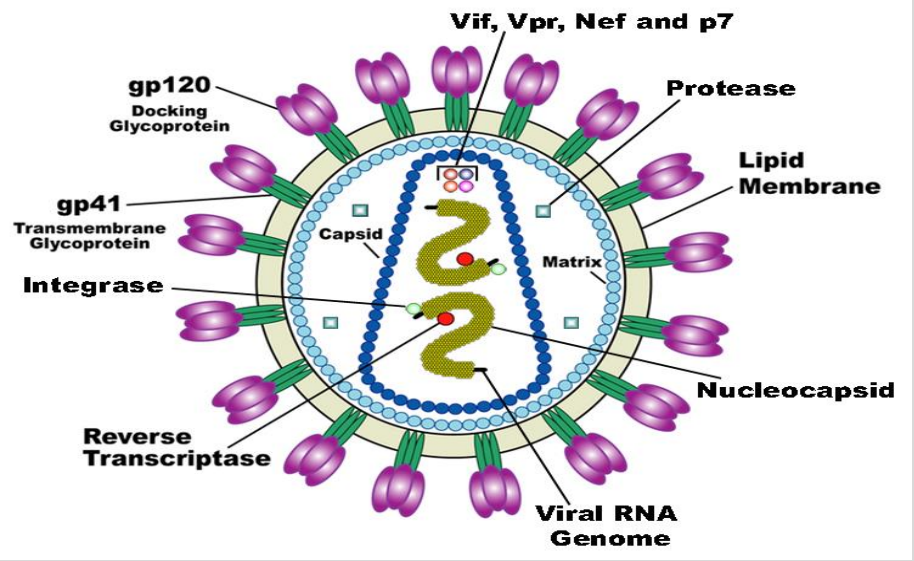
**Human Immunodeficiency Virus (HIV):**

**Classification:**

HIV is a member of the genus ***Lentivirus***, part of the family of **Retroviridae**.

Two types of HIV have been

characterized: HIV-1 and HIV-2.



# Transmission of HIV:

## *HIV could be transmitted by the following routes:*

1- Blood transfusion: (Estimated infections (risk) per 10,000 exposures to an infected source= 9,000 (90%)[1] ). New estimated rate in Canada=1 per 1.3 million donation.

2- Mother-to-child, including pregnancy, childbirth and breastfeeding: (Estimated infections(risk) per 10,000 exposures to an infected source= 2,500 (25%)[2] ).

3- Sexual transmission: (Estimated risk from 0.11-1.7%)[3,4].

4- Needle-sharing injection drug use: Estimated risk=67 (0.67%)[5] .

5- Percutaneous needle stick: Estimated risk= 30 (0.30%)[6] .

6- Tissue transplantation.

# Human T-Lymphotropic Viruses (HTLV) type I and type II:

HTLV is known to cause **adult T-cell leukemia** and lymphoma.

The lymphokine activity places the infected cell in an uncontrolled autocrine mode of growth.

- Classification: Retrovirusvirus; subfamily: *Deltaretrovirus.*

# Transmission of HTLV:

*HTLV-I appears to be transmitted by:*

1- Sexual route.

2- Blood transfusion.

3- Vertical transmission.

4- Breast-feeding.

**Epidemiology:**

It is endemic in Southern Japan (15-30%), Caribbean (3-6%), Guinea and some parts of Africa.

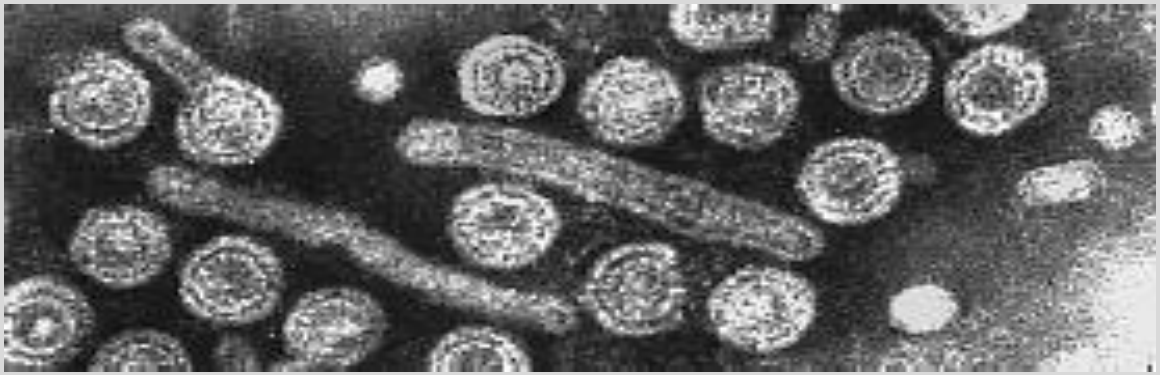
**Laboratory diagnosis:**

The risk has been estimated to be 1 in 1.3 million donations in Canada (1), following the implementation of **PCR** for detection of provirus genome.

Anti-HTLV antibodies could be detected by **ELISA**.

# Hepatitis B Virus:

- The virus is one of the smallest enveloped animal viruses, with a virion diameter of 42 nm.



## Transmission:

1- Sexual Route.

2- Vertical Route: Transmission of virus from mother to child during childbirth.

3- Parenteral Route: Blood transfusion and contaminated syringes.

In Canada, It has been estimated that the risk was 1 in 89,000 donations and came primarily from donations collected during the window period

**Diagnosis**: HBsAg test by ELISA.

anti-HBcAg IgM antibodies ELISA.

# Hepatitis C virus:

## Virology:

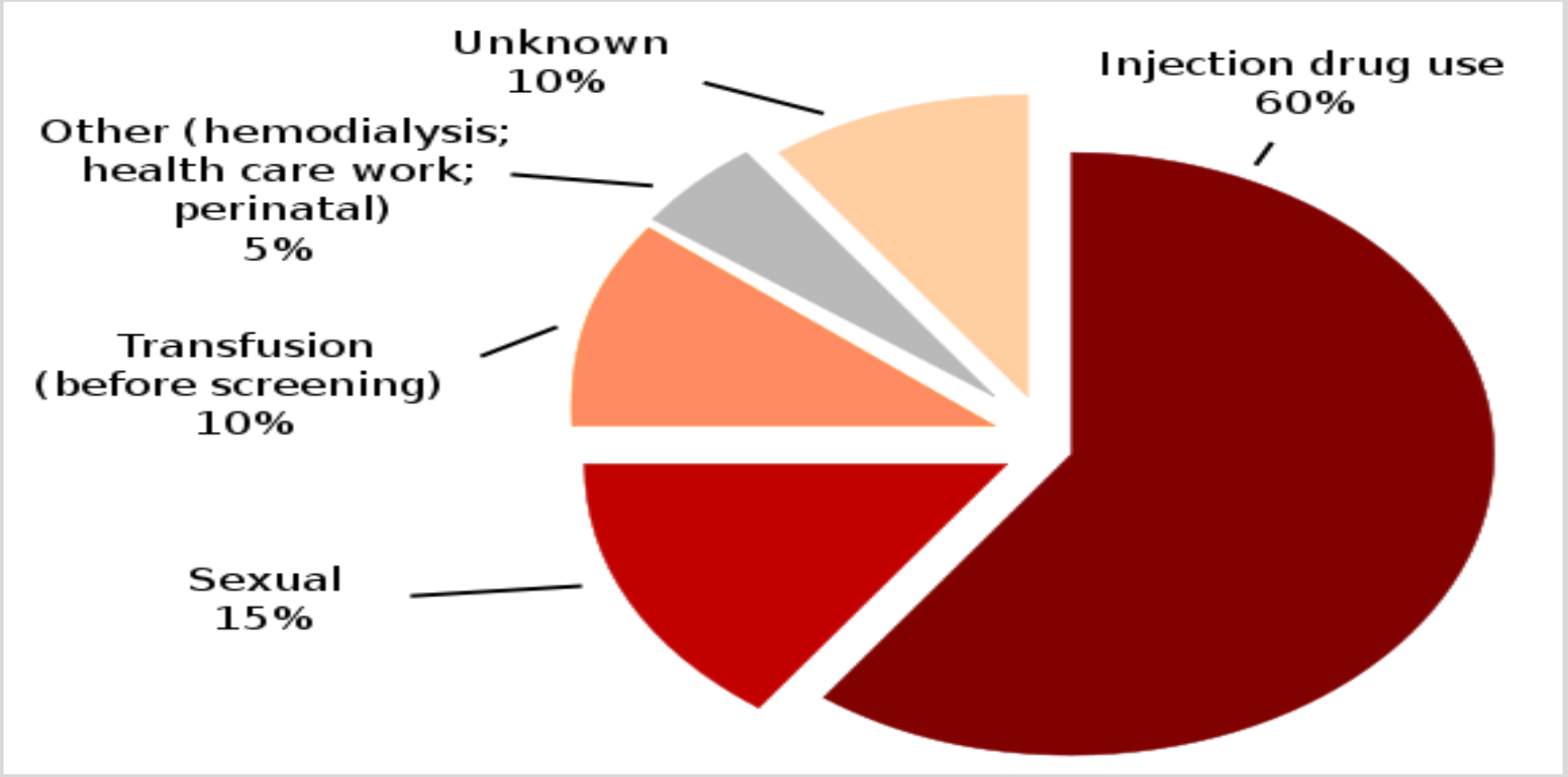
- Small (50 nm in size), enveloped, single-stranded, positive sense RNA virus.

## Transmission:

***Blood-to-blood contact:***

In developed countries, it is estimated that 90% of persons with chronic HCV infection were infected through transfusion of unscreened blood or blood products or via injecting drug use or sexual exposure

In developing countries, the primary sources of HCV infection are unsterilized injection equipment and infusion of inadequately screened blood and blood products.




**Hepatitis C transmission in developed countries.**

# Epstein - Barr virus: (E.B Virus):

**Transmission:**

1-Direct person-person contact (Saliva).

2- Blood transfusion: 90 % of the adult population is seropositive.

E.B virus infects nasopharyngeal epithelial cells, salivary and **lymphoid** tissues.

The virus binds to CD21 of B-Lymphocytes and acts as mitogen.

This will stimulate the production of atypical reactive T cell (**Downey** cells=**70%** of the total WBC count).

# E.B virus Diseases and malignancies:

**Diseases :**

1- Heterophile positive mononucleosis (kissing disease):

fatigue, fever, sore throat, lymphadenopathy, and splenomegaly.

2- Lymphoproliferative disease:

uncontrolled B-Cell growth in immunocompromised patients.

**Malignancies:**

1- Burkitt lymphoma.

2- Nasopharyngeal carcinoma.

3- Hodgkin lymphoma.

**Diagnosis (serology):**

Anti-E.B virus IgM antibodies examined by ELISA test.

# Cytomegalovirus (CMV):

**Virology**: belongs to the *Herpesviridae* family such as E.B. Virus.

**Transmission**: Saliva, sexual, Parenteral, in Utero.

**Pathogenesis:**

-CMV infects the salivary gland epithelial cells, and establishes a persistent infection in fibroblasts, epithelial cells, and macrophages.

-Latency in mononuclear cells.

-Approximately **50% to 80%** of the adult population are infected with the virus.

**Disease**:( in immunocompromised individuals)

1- Cytomegalic inclusion disease.
   jaundice, hepatosplenomegaly, pneumonitis, CNS damage to death.

2- Mononucleosis.

3- Interstitial pneumonitis to severe systemic infection (immunocompetent pateints : AIDS, Tissue transplant).

# Parvovirus B19:

**Transmission:**

respiratory, vertical (from mother to child), blood transfusion.

This virus has been transmitted to patients with hemophilia through infusion of clotting factors (factor VIII and factor IX)[1] .

**Pathogenesis:**

It infects immature erythroid progenitor cells, resulting in cell lysis.

The resulting anemia is clinically significant in patients with sickle cell anemia.

# Human Herpes Virus 8 (HHV-8):

**Transmission**:

Sexual contact, Saliva, vertical, transplantation.

**Pathogenesis:**

HHV-8 has a gene that turns on vascular endothelial growth factor (VEGF), which plays a direct role in the development of Kaposi Sarcoma.

**Disease**:

Kaposi Sarcoma.

**Diagnosis:**

1- Serology –ELISA.

2- Molecular genetics : PCR.

# The Bacterial contamination:

Transfusion Transmitted bacterial reaction has been identified as the most common and severe infectious complication associated with transfusion.

Approximately **57%** of all Transfusion Transmitted infections and 16% of transfusion-related deaths have been associated with bacterial contamination (1).

It has been estimated that 1 in 38,500 units of red cells, 1 in 3,300 units of random donor platelets, are contaminated with bacteria (1).

# *Treponema pallidium:*

**Transmission**: sexually, transplacental, vertical, blood transfusion (Rare; 6 in million blood donation (1)).

**Bacteriology:**

-Thin spirochete; Basically it has Gram's negative cell envelope.

-Axial filaments present (endoflagella).

-Can not be cultivated in vitro; sero-diagnosis.

-Obligate pathogen

(but not intracellular).

**Pathogenesis:**

- Primary:

  genital chancre of skin.

- Tertiary:

  CNS and cardiovascular infection.



## Diagnosis:

**1- Direct**: examination of microbe by Darkfield or immunofluorescent microscopy.

**2- Indirect**: **Serology**:

A-Nontreponemal antibodies (non-specific): looking for Anti-Cardiolipin antibodies in patient serum by VDRL or RPR test.

Venereal disease research laboratory or RPR Rapid plasma reagin (RPR) tests.

B- Treponemal antibodies (specific): looking for anti-treponema antibodies by TPHA or FTA.

- T. pallidum haemagglutination assay (TPHA).
- Flurescent treponemal antibody (FTA).

# Malaria:

It has been estimated that about 300-500 million people are infected, and over 1 million people die from this disease each year.

**Transmission:**

Horizontal (bite of infected female *Anopheles* mosquito), Transplacental, and Blood transfusion (asymptomatic donors).

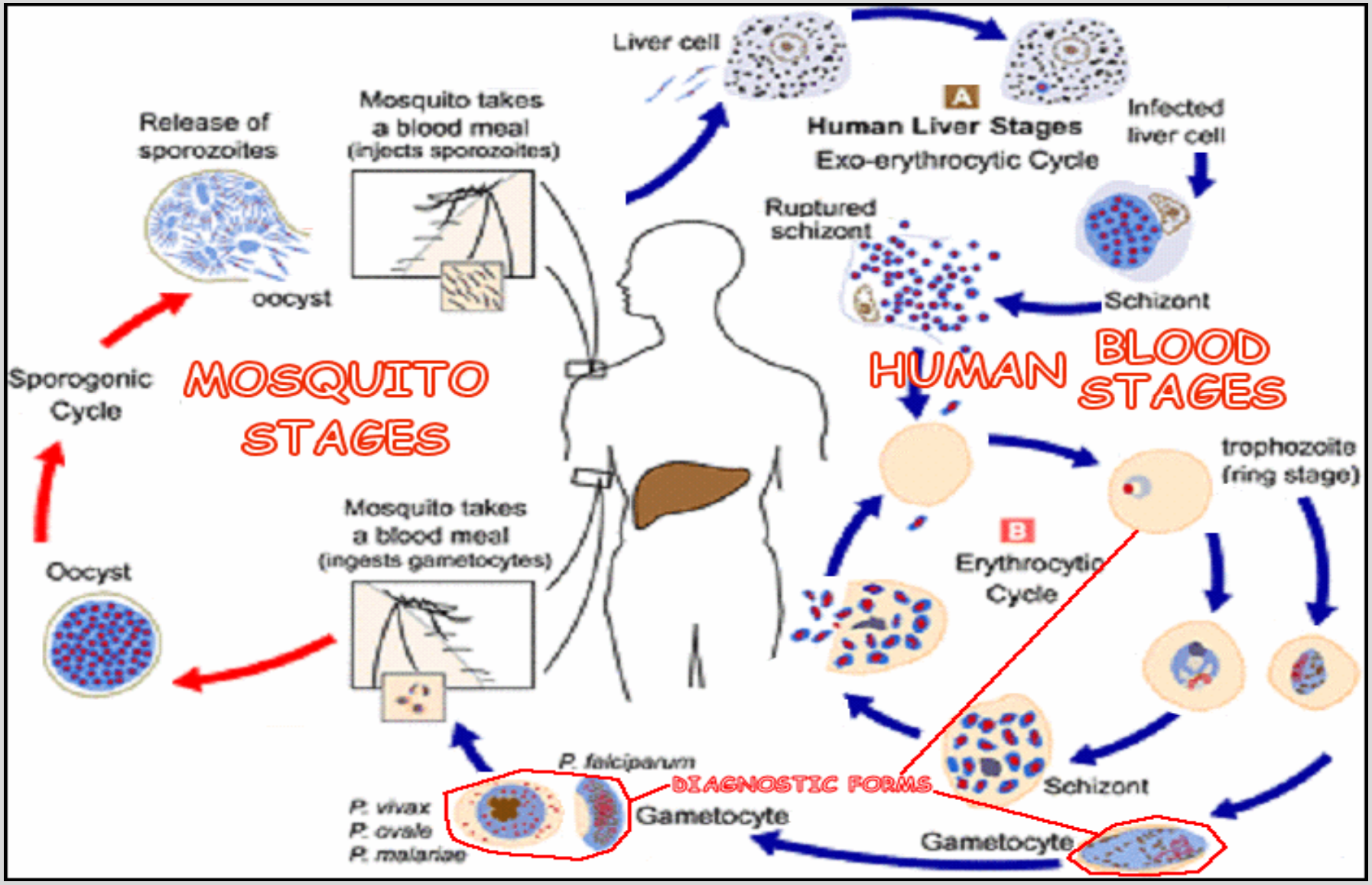
**Pathogenesis:**

1- Infection of liver: jaundice : Hepatomegaly, splenomegaly.

2- Infection of Red blood cells: infectious hemolytic anemia.

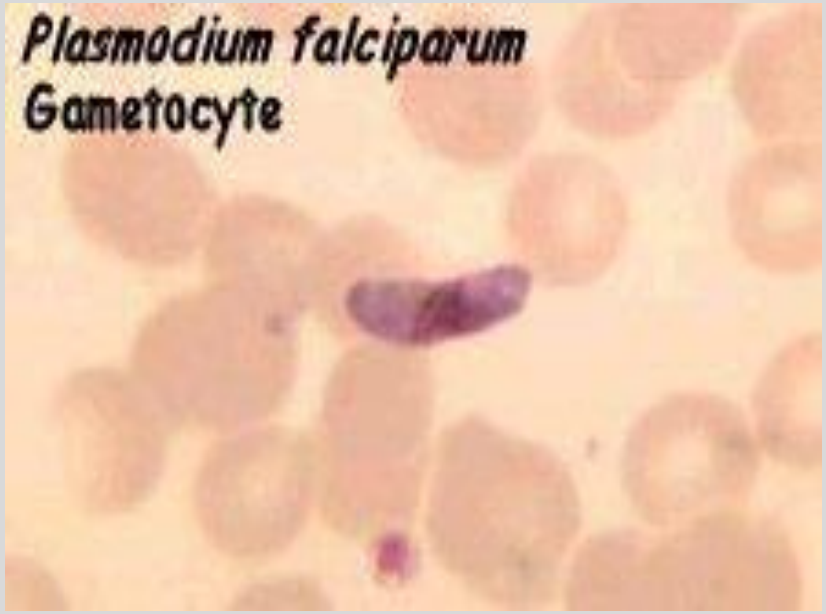
# *Plasmodium* life cycle:

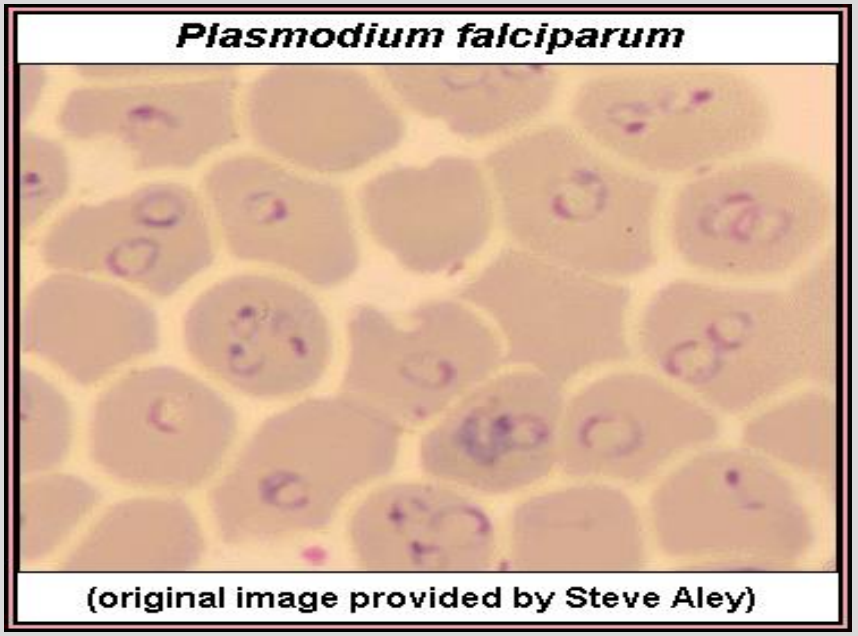
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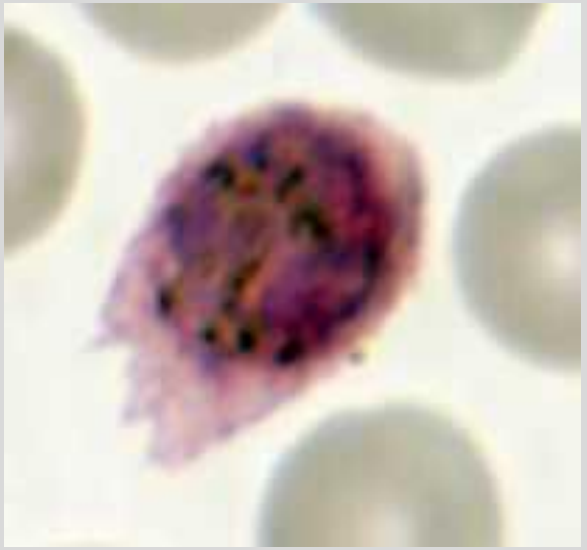
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Plasmodium falciparum
Gametocyte


Plasmodium falciparum
(original image provided by Steve Aley)







# Oncology Cancer Nursing



# General consideration

- Cancer can be considered a chronic disease requiring ongoing management, rather than a terminal illness

- characterized by uncontrolled growth and spread of abnormal cells.

- The malignant cell is able to invade the surrounding tissue and regional lymph nodes

- Metastasis is the secondary growth of the primary cancer in another organ (secondary).

- Most patients die as a result of metastases rather than progression of the primary cancer.



# ETIOLOGY

- Globally, cancer claims an estimated 7.6 million lives, with more than 12 million people receiving cancer diagnoses in a year.
- Cancer is the second leading cause of death in the United States,
- Age is the most outstanding risk factor for cancer. Cancer incidence increases progressively with age.
- Approximately 77% of people diagnosed with cancer are over age 55.
- Two-thirds of all cancers are related to lifestyle habits (eg, smoking, alcohol consumption, diet, physical activity) and environmental carcinogens.



- Alcohol is associated with an increased risk of breast cancer due to alcohol-induced increases in circulating estrogens.
- Exposure to carcinogens, such as asbestos, benzene, and radiation, increases the risk of developing certain types of cancer.
- Solar ultraviolet radiation exposure increase risk of skin cancers.
- There is a hereditary predisposition to specific forms of cancers are linked to certain events within a gene (eg, BRCA1 and BRCA2 in breast cancer, MSH in colon cancer).
- Infections and viruses are risk of certain forms of cancer:
  - a. Human papilloma virus (HPV)—cervical cancer, tonsillar cancer
  - b. Epstein-Barr virus—lymphoma
  - Patterns of incidence and death rates vary with sex,
  - age, race, and geographic location



# Benign v. Malignant

- Benign: overgrowth of cells that are NOT cancerous

- Malignant: overgrowth of cells that ARE cancerous



# Grade Vs stage

- The grade of a tumor is an indication of how quickly it is likely to grow and spread, the grade refers to the appearance of the cancer cells under the microscope.

- The stage of a solid tumor refers to the extent of the tumor, lymph node involvement, and whether or not it has spread to other organs and tissues. The actual number used to describe a tumor stage – 0, 1, 2, 3, or 4



# Detection and Prevention

***Primary Prevention***

The assessment or reduction of risk factors before the disease occurs:

- Make appropriate lifestyle changes.
- Stop smoking.
- Limit alcohol intake.
- Eat a healthy diet as outlined above.
- Be physically active: maintain a healthy weight



# *Primary Prevention*

- Avoid sun exposure, especially during the hours of 10 A.M. and 4 P.M. and cover exposed skin.
- Those at high risk for certain cancers should consider genetic counseling and testing.
- Chemoprevention.

a. Aspirin—low-dose aspirin may reduce risk colon cancer in patient with a history of polyps.

b. Tamoxifen can reduce the risk of breast cancer in women who are at high risk by nearly 50%.

c. Finasteride may reduce the risk of prostate cancer.

d. COX-2 inhibitors—may reduce the risk of GI cancers.

e. Vitamin D—may play a role in reducing the risk of breast cancer.

f. Statins may reduce colon, breast, and hematologic malignancies.



# *Primary Prevention*

- Vaccinations—HPV is now known to cause about 70% of cervical cancer. HPV vaccines aimed at preventing genital warts, precancerous cervical lesions, and cervical cancer are now available. Routine HPV vaccination is recommended for females ages 11 to 12.
- Screening for cervical cancer should continue in both vaccinated and unvaccinated women

***Secondary Prevention***

Screening and early detection to improve overall outcome and survival:

- Performing routine screening tests
- Breast cancer is the most common type of cancer in women
- Colon cancer screening should begin for all men and women over age 50.



# Diagnostic Evaluation

- Complete medical history and physical examination.
- Biopsy of tumor site to determine pathologic diagnosis.

a. The malignancy is classified according to anatomic extent and histopathologic analysis.

b. Biopsy is obtained from the most accessible site (eg, lymph node).

- The grade of the tumor (rating of 1 to 4) is based on how well differentiated the tissue or cells appear.
- Flow cytometry testing of tumor tissue determines the deoxyribonucleic acid (DNA) content



# Diagnostic Evaluation

- Estrogen and progesterone levels are obtained from breast and possibly ovarian tissue.
- Laboratory tests (CBC and blood chemistry).

a. Further testing depends on cancer diagnosis.

b. Blood markers (carcinoembryonic antigen, PSA, CA15-3, CA125)

- Imaging procedures—X-ray, CTscan, MRI and positron-emission tomography to determine extent of metastasis.
- Staging is necessary at the time of diagnosis to determine extent of disease to determine prognosis and management.



**Diagnosis**

1. Biopsy: Tissue is necessary to make a diagnosis to determine the histology and/or grade of a tumor
2. Fine-needle aspiration (FNA) is a technique in which cells are aspirated from a tumor using a needle and syringe.
3. Core-needle biopsy is the simplest method of diagnosis. is inexpensive and causes minimal disturbance of the surrounding tissue, retrieves a small piece of intact tumor tissue, which yields enough tissue
4. Open biopsy is done in the operating room, is more expensive, and requires a longer period of recovery.



# M A N A G E M E N T

- The method of treatment depends on the type of **malignancy**, the specific **histologic cell type**, **stage**, **presence of metastasis**, and **condition of the patient**.
- The four modalities of treatment are:

surgery, chemotherapy, radiation therapy, and biologic therapy or a combination of these modalities.



# Treatment

Surgical treatment of cancer focuses on primary treatment, adjuvant therapy, salvage treatment, and palliative treatment.

1. Primary treatment—involves the removal of a malignant tumor and a margin of adjacent normal tissue.

a. Local excision—simple excision of a tumor and a small margin of normal tissue.

b. Wide excision—includes the removal of the primary tumor, regional lymph nodes, and neighboring structures.

2. Adjuvant treatment—involves the removal of tissues to decrease the risk of cancer recurrence.

a. Debulking surgery is the removal of the bulk of the tumor; should be performed before the start of chemotherapy.



3. Salvage treatment—involves the use of an extensive surgical approach to treat a local recurrence

4. Palliative treatment—attempts to relieve the complications

of cancer (eg, obstruction of the GI tract, pain).

5. Reconstructive/rehabilitative surgery—repair of defects from previous radical surgical resection (breast reconstruction)

6. Preventive/prophylactic surgery—removal of lesions that, if left in the body, are at risk of developing into cancer



# CHEMOTHERAPY

- Chemotherapy is the use of antineoplastic drugs to promote tumor cell destruction by interfering with cellular function and reproduction.
- It includes the use of various chemotherapeutic agents and hormones.
- The intent of chemotherapy is to destroy as many tumor cells as possible with minimal effect on healthy cells.

**Chemotherapy is utilized in different clinical settings:**

- As induction chemotherapy for advanced disease /no alternative treatment exists.
- As an adjunct to local methods of treatment.
- Adjuvant chemotherapy is the use of systemic treatment after the primary tumor has been controlled by surgery and/or radiation therapy.
- To palliate symptoms of metastatic disease and/or prolong survival.



# CHEMOTHERAPY

- Chemotherapeutic agents can be effective on one of the four phases of the cell cycle:

  a. G1 (gap one) phase: RNA and protein synthesis (enzymes for DNA).

  b. S (synthesis) phase: the DNA component doubles for the chromosomes in preparation for cell division.

  c. G2 (gap two) phase: protein and RNA synthesis occurs, and the mitotic spindle apparatus is formed.

  d. M (mitosis) phase: the cell divides into two identical daughter cells.

  e. Cells not active in the cell cycle are designated as “resting” (G0). Cells in this phase are, for the most part, refractory to chemotherapy.



# Routes of administration:

a. Oral—capsule, tablet, or liquid

b. I.V.—push (bolus) or infusion over a specified time period.

c. Intramuscular

d. Intrathecal/intraventricular

e. Intra-arterial

f. Intracavitary—such as peritoneal cavity

g. Intravesical—into uterus or bladder

h. Topical

Chemotherapy predictably affects normal, rapidly growing cells (eg, bone marrow, GI tract lining, hair follicles).



# Adverse Effects of Chemotherapy

All chemotherapy drugs have adverse effects especially patient younger than age 1 or is elderly, has co-morbid conditions, or has impaired renal or hepatic function.

- ***Alopecia***

1. Most chemotherapeutic agents cause some degree of alopecia. This is dependent on the drug dose, half-life of drug, and duration of therapy.

2. Usually begins 2 weeks after administration of chemotherapy.

Regrowth takes about 3 to 5 months.

3. The use of scalp hypothermia and tourniquets is highly controversial.

- ***Anorexia***

1. Chemotherapy changes the reproduction of taste buds.

2. Absent or altered taste can lead to a decreased food intake.

3. Concurrent renal or hepatic disease can increase anorexia.



## *Mucositis*

1. Caused by the destruction of the oral mucosa
2. Consistent oral hygiene is important to avoid infection.

## *Myelosuppression (*Neutropenia)

1. An absolute neutrophil count (ANC) of 1,000/mm3 or less.
2. Risk of infection is greatest with an ANC less than 500/mm3.
3. Caused by suppression of the stem cell.
4. Usually occurs 7 to 14 days after administration of chemotherapy
5. Patients should be taught to proper hand washing, avoiding those with illness, and proper hygiene, eliminating raw or unwashed meat and fish
6. Monitor promptly for fever or other signs of infection.
7. Can be prevented with the use of growth factors (eg, granulocyte colony-stimulating factor) such as Lenograstim.



- ***Diarrhea***

  1. can lead to severe dehydration, electrolyte imbalances, hospitalizations, and treatment delays and up to 90% of all patients experience diarrhea.

- ***Fatigue***

  The cause of fatigue is generally unknown but can be related to anemia, weight loss, altered sleep patterns, and coping.

- ***Nausea and Vomiting***

  1. Caused by the stimulation of the vagus nerve by serotonin released by cells in the upper GI tract.

  2. Incidence depends upon the particular chemotherapeutic agent and dosage.

  3. Patterns of nausea and vomiting:

     *Acute*—occurs 0 to 24 hours after chemotherapy administration.

     *Delayed*—can occur 1 to 6 days after chemotherapy administration



**Anemia**

1. Caused by suppression of the stem cell or interference with cell proliferation pathways.

2. May require red blood cell transfusion or injection of erythropoietin.

**Thrombocytopenia**

1. Caused by suppression of megakaryocytes.

2. Risk of bleeding is present when platelet below 50,000/mm3.

4. Risk is high when count falls below 20,000/mm3 (May require platelet transfusions).

5. Risk is critical when count falls below 10,000/mm3.

6. Patient should be taught to avoid injury (eg, no razors), vaginal douches and rectal suppositories, and dental floss during the period of thrombocytopenia.

***Hypersensitivity Reactions***



# RADIATION THERAPY

- Radiation therapy is the treatment of benign and malignant diseases with ionizing radiation.
- The goal of radiation is to deliver a precisely measured dose of irradiation to a defined tumor volume with minimal damage to surrounding healthy tissue.
- This results in eradication of tumor, high quality of life, prolongation of survival, and allows for effective palliation or prevention of symptoms of cancer, with minimal morbidity.
- Improved targeting and increased sparing of normal tissues.

- ***Goals of Therapy***

1. Curative—when there is a probability of long-term survival
2. Palliative—when there is no hope of survival



***Types of Radiation Therapy***

- External (teletherapy) and internal (brachytherapy).

1. Teletherapy is commonly delivered through a linear accelerator machine

    a. Intraoperative radiation therapy (IORT)—given during surgery

    b. Stereotactic radiation (radiosurgery)—used for brain tumors, this can be administered through gamma knife technology.

    c. Intensity-modulated radiation therapy (IMRT)—newer technique that uses radiation beams of varying intensities to deliver different doses to small areas of tissue at the same time(brain, head, and neck tumors and lung tumors).

2. Brachytherapy uses seeds, wires, or catheters to deliver radiation directly (breast and prostate cancers, peritoneal)

- **BIOLOGIC THERAPY**

The goal is to produce antitumor effects through the action of natural host

- defense mechanisms. It is capable of altering the immune system with either stimulatory or suppressive effects



# SPECIAL CONSIDERATIONS IN CANCE RCARE

- Pain R/T cancer may be caused by direct tumor infiltration of bones, nerves, viscera, or soft tissue or by prior therapeutic measures (surgery, radiation).
- The incidence from 14% to 100% of all patients with cancer.

**Types of Pain**

- 1. Somatic pain—caused by direct tumor involvement of sensory receptors in cutaneous and deep tissues

    a. dull, sharp, aching, and throbbing; usually constant and localized.

    b. somatic pain is bone pain caused by metastasis.

    c. Can usually be controlled with NSAIDs or oral opioids.



# Types of Pain-cont.

- 2. Neuropathic pain

    a. Results from nerve injury or compression.

    b. Includes phantom pain and postherpetic neuralgia.

    c. Described as burning, shooting, electric, and vlancinating. (It can be constant or sporadic.)

    d. Usually associated with paresthesia.

    e. Treatment usually includes combinations of antidepressants, anticonvulsants, and opioids.

- 3. Visceral pain

    a. Usually described as deep, dull, aching, squeezing, or pressure sensation.

    b. Usually caused by abnormal stretching of smooth muscle walls, ischemia of visceral muscle, and serosal irritation.

    c. Can be treated with oral opioids or surgery to remove the cause.

Pharmacologic Management

1. ***Nonopioid analgesics***

- Acetaminophen, NSAID, Corticosteroids, Bisphosphonates (used for lytic bone lesions to prevent fractures and may help control bone pain), Anticonvulsants and antidepressants (neuropathic pain)

***2. Opioid analgesics***

**ONCOLOGIC EMERGENCIES**

**a. Septic Shock b. Spinal Cord Compression c. Hypercalcemia**

- **PALLIATIVE CARE**

is the active total care of patients with advanced illness, is no longer on curative treatment, but on quality of life and integrating the physical, psychological, spiritual, and social aspects of care.

- **End-of-Life Care**

Hospice and palliative care provide care during progressive illness, but hospice eligibility begins only when a patient has a life expectancy of 6 months or less.

# Integumentary Health

# Primary Lesions

1. Macule: flat, discoloration of skin; have any size or shape.
2. Papule: solid, elevated lesion less than 1 cm wide.
3. Nodule: raised, solid lesion larger than 1 cm wide.
4. Vesicle: elevated lesion less than 1cm, containing fluid.
5. **Bulla: a vesicle or blister larger than 1 cm wide.**
6. **Pustule: circumscribed raised lesion that contains pus**
7. **Wheal: elevation of the skin that lasts less than 24 hours, caused by edema of the dermis**

8. Plaque: solid, elevated lesion on the skin or mucous membrane, larger than 1 cm in diameter; psoriasis is commonly manifested as plaques on the skin; leukoplakia is an example of plaques on mucous membranes.

9. Cyst: soft or firm mass in the skin, filled with semisolid or with liquid material contained in a sac.

10. Petechiae: deposits of blood or blood pigment less than 1 cm wide.

11. Purpura—deposits of blood or blood pigment greater than 1 cm

# Secondary Lesions

involve changes that take place in primary lesions

1. Scale: heaped-up, horny layer of dead epidermis; may develop as a result of inflammatory changes.
2. Crust: covering formed by the drying of serum, blood, or pus on the skin.
3. Excoriation—linear scratch marks or traumatized areas of skin.
4. Fissure—cracks in the skin
5. Ulcer—lesion formed by local destruction of the epidermis
6. Lichenification—thickening of skin accompanied by accentuation of skin markings.

7. Scar: new formation of connective tissue that replaces the loss of substance in the dermis.

8. Atrophy: diminution in size or in loss of skin cells

9. Keloid: hypertrophic scar that is larger than the original lesion or injury.

- Skin Graft: A section of skin tissue is separated from its blood supply and transferred as free tissue to recipient

a. Autografts: grafts tissue transplanted from the pt own skin.
b. Allografts (homografts): tissue from one individual.
c. Xenograft (heterograft): the transfer of tissue from another species.

**CELLULITIS**

- *Cellulitis* is a diffuse inflammation of the deep dermal and subcutaneous tissue that results from an infectious process.

**Clinical Manifestations**

1. Tender, warm, erythematous, and swollen area
2. Possible fluctuant abscess or purulent drainage.
3. Possible fever, chills, headache, malaise.

**Diagnostic Evaluation**

1. Gram stain and culture of drainage.
2. Blood cultures

**Management (**penicillinase-cephalosporins, quinolones, or sulfa-based, Parenteral antibiotics, Surgical drainage)

- **Complications (**Tissue necrosis & Septicemia)

**PSORIASIS**

- is a chronic T-cell mediated inflammatory disorder causing secondary epidermal turnover, which occurs at a rate 6 to 9 times faster than normal.

**Pathophysiology and Etiology**

- Affects 1% to 3% of people.
- Formerly considered idiopathic and immune system modulated.
- Condition tends to last a person's life span, with flare-ups and remissions. May be exacerbated by infection; drugs,

Types of psoriasis include:

a. Plaque: most common type (knees, elbows, scalp).

b. Guttate: occurs on trunk, arms, legs; triggered streptococcal infection.

c. Pustular—blisters contain puslike material on hand and feet.

f. Psoriatic arthritis—arthritis .

**Clinical Manifestations**

- Erythematous plaques with silvery scales.
- Symmetric involvement.
- May be pruritic and painful.
- Often appear at areas of epidermal injury reaction.
- Characteristic pitting of nails in 50% of patients.
- Arthritis in approximately 10% of patients.

**Management**

- Coal tar and anthralin (antimitotic) preparations inhibit excessive skin turnover.
- Topical corticosteroids.
- Tachyphylaxis may result if used for extended periods.
- Topical calcipotriene (Dovonex), a vitamin D derivative,
- tazarotene (Tazorac), a receptor selective retinoid (pregnancy category X).
- Phototherapy—20 to 30 short treatments and UV light
- Oral methotrexate, the retinoid acitretin (Soriatane), cyclosporine
- Biologic agents, such as etanercept (Enbrel), infliximab

# Scabies Transmission



- Direct, prolonged, skin–to-skin contact
- Sexual contact
- Household contact
- Indirect transfer from clothing, towels and bedding, only if contaminated immediately before contact
- Communicable
  - while infested and untreated
  - during incubation period



# Scabies Symptoms

- Symptoms caused by reaction to mites
- Pimple-like rash or burrows between fingers, on wrist, elbows, armpits, belt line, navel, abdomen, and/or buttocks
- Intense itching over most of the body, especially at night
- Sores on the body caused by scratching
- Incubation period:
  - 2-6 weeks without previous exposure
  - 1-4 days after re-infestation (usually milder)

# Scabies Treatment

- Applicaton of scabicide over entire body below head:
  - 5% permethrin cream (wash after 8-14 hrs)
  - 1% lindane cream (wash after 8-12 hrs)
- Ivermectin, single dose, orally
- Itching may continue for several weeks despite successful treatment
- In ~5% of cases, 2$^{nd}$ treatment may be necessary after 7-10 days.

# Applying Prescription Creams

- Follow package insert
- Apply thin layer over whole body from neck down, paying particular attention to folds in skin
- Trim fingernails and toenails, apply under nails
- Reapply after handwashing
- Leave on for at least 8 hours, but not longer than maximum time recommended for product
- Wash off with warm, soapy water and dry skin

# Scabies Control

- Prompt diagnosis and treatment of patients
- Simultaneous, prophylactic therapy:
  - household members,
  - sexual contacts, and
  - caregivers with prolonged skin-to skin contact
- Launder all clothing worn within 48-72 hrs. of treatment in hot water and dry in hot dryer
- Thorough vacuuming of carpets and upholstered furniture

Varicella (Chickenpox)

- is a disease caused by the varicella zoster virus; it causes fever and an itchy rash
- Varicella-zoster virus (VZV) causes two diseases: •varicella (chickenpox), the primary infection, •and herpes zoster, a secondary infection due to a reactivation of latent varicella infection in the dorsal root ganglia Infectious agent
- First fever, headache, loss of appetite then a skin rash covering the body but mostly concentrated in the face, scalp and trunk.
- At lease 90% of the population has had chickenpox by age 15
- It is spread through: saliva, coughing, sneezing, contact with blisters
- Incubation period •10-21 days •Commonly 14-16 days
- Antihistamine or topical ointments to relieve itching.
- You can also soothe itching skin by:
- taking lukewarm baths, applying unscented lotion
- wearing lightweight, soft clothing



**Burn**

- Burns are a form of traumatic injury caused by thermal, electrical, chemical, or radioactive agents.
- Inhalation injury and associated pulmonary complications are a significant factor in mortality and morbidity from burn injury (50% to 60% of fire deaths are secondary to inhalation injury).
- The very young and the elderly are at greatest risk for burn injuries.
- Males are more commonly injured by burns than are females.

- Physiologic reaction to a burn is similar to the inflammatory process.

a. Adjacent intact vessels dilate, causing redness and blanching pressure.

b. Platelets and leukocytes begin to adhere to the vascular endothelium

c. Increased capillary permeability produces wound edema.

d. Eventually, new capillaries, immature fibroblasts, and newly formed collagen fibrils appear within the wound.

- Burns may be partial (second degree) or full (third degree) thickness
- Burn depth is directly related to the temperature of the burning agent and the duration of contact with body tissue.

a. Below 44.4° C, no local damage occurs

b. At 48.9° C, it takes 5 minutes' exposure to create a full-thickness burn.

c. At 51.7° C the time requirement is 2 minutes, and at 60° C only 6 seconds required.

d. At 70.6° C it takes 1 second to create a full thickness burn.

## nhalation Injury

1. May be upper airway (supraglottic) and cause ARDS.

2. Carbon monoxide is a colorless, odorless, tasteless, nonirritating gas

3. Affinity of hemoglobin for CO is 200 times greater than for oxygen.

4. Toxicity depends on concentration of carbon monoxide in inspired air

and the length of time of exposure.

- Levels of 20% to 50% can produce headache, fatigue, irritability, cardiac dysrhythmias, ataxia, vomiting, syncope, tinnitus, dystonia, impaired judgment, loss of consciousness, deep coma, hypotension, seizures, respiratory arrest, and death.
- Sulfur dioxide and nitrous oxide are toxic agents in soot.
- Toxic fumes from burning plastic are more dangerous than smoke.
- Noxious gases include hydrogen cyanide, hydrochloric acid, sulfuric acid, halogens, and perhaps phosgene.

# Systemic Changes in Major Burns

1. ***Fluid Shifts*** *(*from vascular to interstitial spaces)

***2. Hemodynamic Changes*** (Lessened circulating blood volume), Hyponatremia due to fluid shift and hyperkalemia initially due to cell destruction, followed by hypokalemia

***3. Metabolic Demands***

***4. Renal Needs*** (Glomerular filtration may be decreased in extensive injury)

- ***5. Pulmonary Changes*** (Initial respiratory alkalosis resulting from hyperventilation may change to respiratory acidosis associated with pulmonary insufficiency)
- ***6. Hematologic Changes*** (Thrombocytopenia, abnormal platelet function)

***7. Immunologic Activity***

***8. GI Impact***

Major burns involving >25% of total body surface

**SEVERITY OF BURNS** is determined by:

1. Depth: first, second (partial-thickness), third degree (full ).

2. Extent—percentage of TBSA.

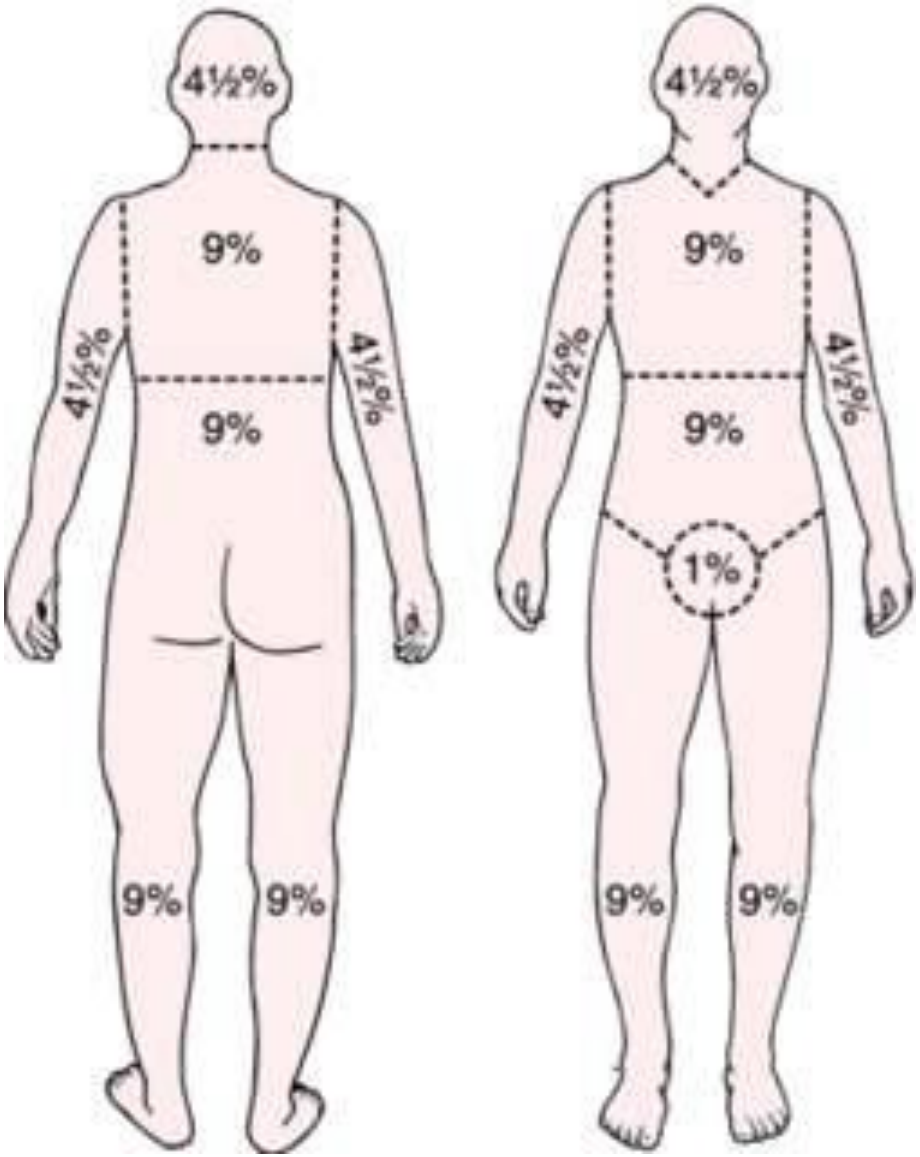
3. Age: the very young and very old have a poor prognosis.

4. Area of the body burned—face, hands, feet, perineum, and circumferential burns require special care.

5. Medical history and concomitant injuries and llness.

6. Inhalation injury.

Rule of nines for calculating total burn surface area (TBSA).



# TABLE 34-2 Assessment of Burn Injury P1177

- Management of acute burn injury includes:
  - hemodynamic stabilization
  - metabolic support wound debridement, use of topical antibacterial therapy **Silver sulfadiazine 1%**, biologic dressings, and wound closure.
  - Prevention and treatment of complications, including infection and pulmonary damage, and rehabilitation are also of major importance.
  - The patient will also require physical and occupational therapy and psychiatric and nutritional support.